

Supplementary Materials

Racial and Ethnic Differences in Breast Cancer Incidence in the United States: A Systematic Review with Quantitative Synthesis

Abbreviations

Abbreviation	Definition
AANHPI	Asian American, Native Hawaiian, and Pacific Islander
AI/AN	American Indian and Alaska Native
ANTR	Alaska Native Tumor Registry
CCR	California Cancer Registry
CI	confidence interval
CiNA	Cancer in North America (NAACCR combined incidence dataset)
HER2	human epidermal growth factor receptor 2
HR	hormone receptor
IHS	Indian Health Service
IRR	incidence rate ratio
JB	Joanna Briggs Institute
NAACCR	North American Association of Central Cancer Registries
NHB	non-Hispanic Black
NHPI	Native Hawaiian and Pacific Islander
NHW	non-Hispanic White
NPCR	National Program of Cancer Registries
PRCDA	Purchased/Referred Care Delivery Area (IHS; formerly Contract Health Service Delivery Area, CHSDA)
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RoB	risk of bias
SE	standard error
SEER	Surveillance, Epidemiology, and End Results Program
SIR	standardized incidence ratio
TNBC	triple-negative breast cancer
USCS	United States Cancer Statistics (NPCR + SEER)

Supplementary Table 1. Final search strategy for each database

Database	Platform	Date	Records	Search string
PubMed/MEDLINE	PubMed	2026-08-07	1,331	("Breast Neoplasms"[Mesh] OR "breast cancer"[tiab] OR "breast carcinoma"[tiab] OR "breast neoplasm*"[tiab]) AND (race[tiab] OR racial[tiab] OR ethnic*[tiab] OR minorit*[tiab] OR disparit*[tiab] OR Black[tiab] OR Hispanic[tiab] OR Latin*[tiab] OR White[tiab] OR Asian[tiab] OR "African American"[tiab] OR "American Indian*"[tiab] OR "Alaska Nativ*"[tiab] OR "Native American*"[tiab] OR indigenous[tiab] OR "Pacific Islander*"[tiab] OR "Native Hawaiian*"[tiab] OR "South Asian*"[tiab] OR "Asian American*"[tiab] OR Chinese[tiab] OR Japanese[tiab] OR Korean[tiab] OR Filipin*[tiab] OR Vietnamese[tiab] OR "Asian Indian*"[tiab] OR "Ethnicity"[Mesh] OR "Minority Groups"[Mesh] OR "Health Status Disparities"[Mesh] OR "Black or African American"[Mesh] OR "Hispanic or Latino"[Mesh] OR "American Indian or Alaska Native"[Mesh] OR "Asian"[Mesh]) AND (incidence[tiab] OR "incidence rate*"[tiab] OR "age-adjusted"[tiab] OR "age-standardized"[tiab] OR "age-standardised"[tiab] OR "Incidence"[Mesh]) AND ("United States"[tiab] OR USA[tiab] OR America*[tiab] OR SEER[tiab] OR NAACCR[tiab] OR USCS[tiab] OR "United States Cancer Statistics"[tiab] OR "United States"[Mesh] OR "SEER Program"[Mesh] OR Connecticut[tiab] OR Iowa[tiab] OR "New Mexico"[tiab] OR Utah[tiab] OR Hawaii[tiab] OR Georgia[tiab] OR Michigan[tiab] OR Washington[tiab] OR California[tiab] OR Kentucky[tiab] OR Louisiana[tiab] OR "New Jersey"[tiab]) AND (2000:2026[dp]) AND English[lang] AND humans[MeSH] NOT (review[pt] OR "case reports"[pt] OR editorial[pt] OR comment[pt] OR letter[pt] OR "news"[pt])
Embase	embase.com (Advanced Search)	2026-08-07	3,248	('breast cancer'/exp OR 'breast cancer' OR 'breast cancer':ti,ab OR 'breast carcinoma':ti,ab OR 'breast neoplasm':ti,ab) AND (race:ti,ab OR racial:ti,ab OR ethnic*:ti,ab OR minorit*:ti,ab OR disparit*:ti,ab OR black:ti,ab OR hispanic:ti,ab OR latin*:ti,ab OR white:ti,ab OR asian:ti,ab OR 'african american':ti,ab OR 'american indian':ti,ab OR 'alaska nativ*':ti,ab OR 'native american*':ti,ab OR indigenous:ti,ab OR 'pacific islander*':ti,ab OR 'native hawaiian*':ti,ab OR 'south asian*':ti,ab OR 'asian american*':ti,ab OR chinese:ti,ab OR japanese:ti,ab OR korean:ti,ab OR filipin*:ti,ab OR vietnamese:ti,ab OR 'asian indian*':ti,ab OR 'ethnic group'/exp OR 'ethnic group' OR 'race'/exp OR 'race' OR 'health disparity'/exp OR 'health disparity' OR 'american indian'/exp OR 'american indian' OR 'hispanic'/exp OR 'hispanic' OR 'ethnicity'/exp OR 'ethnicity') AND (incidence:ti,ab OR 'incidence rate':ti,ab OR 'age-adjusted':ti,ab OR 'age-standardized':ti,ab OR 'age standardization') AND ('united states':ti,ab OR usa:ti,ab OR america*:ti,ab OR seer:ti,ab OR naaccr:ti,ab OR uscs:ti,ab OR 'united states cancer statistics':ti,ab OR 'united states'/exp OR 'united states' OR connecticut:ti,ab OR iowa:ti,ab OR 'new mexico':ti,ab OR utah:ti,ab OR hawaii:ti,ab OR georgia:ti,ab OR michigan:ti,ab OR washington:ti,ab OR california:ti,ab OR kentucky:ti,ab OR louisiana:ti,ab OR 'new jersey':ti,ab) AND [2000-2026]/py AND [english]/lim AND [humans]/lim NOT ('review'/it OR 'case report'/it OR 'editorial'/it OR 'note'/it OR 'conference abstract'/it)
Scopus	scopus.com (Advanced Search)	2026-08-07	2,438	(TITLE-ABS-KEY ("breast cancer" OR "breast carcinoma" OR "breast neoplasm*")) AND (TITLE-ABS-KEY (race OR racial OR ethnic* OR minorit* OR disparit* OR black OR hispanic OR latin* OR white OR asian OR "african american" OR "american indian*" OR "alaska nativ*" OR "native american*" OR indigenous OR "pacific islander*" OR "native hawaiian*" OR "south asian*" OR "asian american*" OR chinese OR japanese OR korean OR filipin* OR vietnamese OR "asian indian*")) AND (TITLE-ABS-KEY (incidence OR "incidence rate*" OR "age-adjusted" OR "age-standardized" OR "age standardi*ation")) AND (TITLE-ABS-KEY ("United States" OR USA OR America* OR SEER OR NAACCR OR USCS OR "United States Cancer Statistics" OR Connecticut OR Iowa OR "New Mexico" OR Utah OR Hawaii OR Georgia OR Michigan OR Washington OR California OR Kentucky OR Louisiana OR "New Jersey"))

Database	Platform	Date	Records	Search string
				AND PUBYEAR > 1999 AND PUBYEAR < 2027 AND (LIMIT-TO (LANGUAGE , "English")) AND NOT (DOCTYPE (re) OR DOCTYPE (le) OR DOCTYPE (ed) OR DOCTYPE (no) OR DOCTYPE (cp))
Web of Science	Web of Science (Advanced)	2026-08-07	2,082	TS=("breast cancer" OR "breast carcinoma" OR "breast neoplasm*") AND TS=(race OR racial OR ethnic* OR minorit* OR disparit* OR Black OR Hispanic OR Latin* OR White OR Asian OR "African American" OR "American Indian*" OR "Alaska Nativ*" OR "Native American*" OR indigenous OR "Pacific Islander*" OR "Native Hawaiian*" OR "South Asian*" OR "Asian American*" OR Chinese OR Japanese OR Korean OR Filipin* OR Vietnamese OR "Asian Indian*") AND TS=(incidence OR "incidence rate*" OR "age-adjusted" OR "age-standardized" OR "age standardi?ation") AND TS=("United States" OR USA OR America* OR SEER OR NAACCR OR USCS OR "United States Cancer Statistics" OR Connecticut OR Iowa OR "New Mexico" OR Utah OR Hawaii OR Georgia OR Michigan OR Washington OR California OR Kentucky OR Louisiana OR "New Jersey") AND PY=(2000-2026) AND LA=(English) NOT DT=(Review OR Letter OR Editorial Material OR Note OR Meeting Abstract)

Note. Search conducted 7 August 2026. Concept blocks combined with AND: breast cancer × race/ethnicity × incidence/age-adjusted rate × United States. Limits: 2000–2026, English, human; document-type exclusions. Total records identified 9,099; duplicate records removed (cross-database) 4,306; unique records screened 4,793.

Supplementary Table 2. Characteristics of included studies (n = 176)

Study (author, year)	Study design	Data source	Role in synthesis	PMID / DOI
Quantitative synthesis — data extracted (n = 58)				
US incidence of breast cancer subtypes defined by joint hormone receptor and HER2 status (Howlander et al., 2014)	Population-based registry/incidence study	SEER (28% US)	Overlap/sensitivity only (12 cells)	PMID 24777111
Breast Cancer Incidence Rates in Ghanaian and US Black Women from 2013 Through 2015 (Davis Lynn et al., 2025)	Population-based registry/incidence study	SEER 17	Representative for 2 cells; overlap for 1	PMID 41082230
Disparities and trends of the incidence and mortality of female-specific cancers in the United States (Zhang et al., 2025)	Population-based registry/incidence study	SEER 22	Overlap/sensitivity only (4 cells)	PMID 41086189
Breast Cancer Incidence Trends in Older US Women by Race, Ethnicity, Geography, and Stage (Lee Argov et al., 2025)	Population-based registry/incidence study	USCS	Representative for 4 cells	PMID 40553473
Invasive breast cancer among Alaska Native women in Alaska (Nash et al., 2019)	Population-based registry/incidence study	Alaska Native Tumor Registry	Representative for 1 cell	PMID 31234738
Cancer incidence in the Hmong in California, 1988-2000 (Mills et al., 2005)	Population-based registry/incidence study	California Cancer Registry	Representative for 1 cell	PMID 16247793
The Alaska Native Tumour Registry: fifty years of cancer surveillance data for Alaska Native people (Nash et al., 2022)	Population-based registry/incidence study	Alaska Native Tumor Registry	Overlap/sensitivity only (1 cell)	PMID 34918619
Age distributions of breast cancer diagnosis and mortality by race and ethnicity in US women (Hendrick et al., 2021)	Population-based registry/incidence study	SEER 21	Overlap/sensitivity only (4 cells)	PMID 34427920
Disparities in breast cancer subtypes among women in the lower Mississippi Delta Region states (Zahnd et al., 2019)	Population-based registry/incidence study	NAACCR (7 Delta states)	Overlap/sensitivity only (9 cells)	PMID 30972520
Invasive breast cancer incidence trends by detailed race/ethnicity and age (Liu et al., 2012)	Population-based registry/incidence study	LA County (CSP/CA-SEER)	Overlap/sensitivity only (4 cells)	PMID 21351091
Quantification of Potential Inequities in Breast Cancer Incidence in New Mexico Through Bayesian Disease Mapping (Zahrieh et al., 2021)	Population-based registry/incidence study	SEER (New Mexico)	Overlap/sensitivity only (1 cell)	PMID 33705303
Breast cancer incidence among American Indian and Alaska Native women: US, 1999-2004 (Wingo et al., 2008)	Population-based registry/incidence study	SEER/NPCR (IHS-linked)	Overlap/sensitivity only (7 cells)	PMID 18720389
Breast Cancer Incidence Among US Women Aged 20 to 49 Years by Race, Stage, and Hormone Receptor Status (Xu et al., 2024)	Population-based registry/incidence study	SEER 17	Overlap/sensitivity only (4 cells)	PMID 38277147
State Variation in Racial and Ethnic Disparities in Incidence of Triple-Negative Breast Cancer among US Women (Sung et al., 2023)	Population-based registry/incidence study	USCS	Representative for 4 cells	PMID 36862439
The high and heterogeneous burden of breast cancer in Hawaii: A unique multiethnic U.S. Population (Loo et al., 2019)	Population-based registry/incidence study	Hawaii Tumor Registry	Representative for 14 cells	PMID 30503975
Trends in Breast Cancer Incidence, by Race, Ethnicity, and Age Among Women Aged ≥20 Years - United States, 1999-2018 (Ellington et al., 2022)	Population-based registry/incidence study	USCS	Representative for 3 cells; overlap for 1	PMID 35025856
Vital signs: Racial disparities in breast cancer severity - United States, 2005-2009 (Cronin et al., 2012)	Population-based registry/incidence study	USCS	Overlap/sensitivity only (1 cell)	PMID 23151952
Breast cancer epidemiology in blacks and whites: Disparities in incidence, mortality, survival rates	Population-based registry/incidence study	SEER 9	Overlap/sensitivity only (1 cell)	PMID 18507200

Study (author, year)	Study design	Data source	Role in synthesis	PMID / DOI
and histology (Baquet et al., 2008)				
Breast Cancer Incidence in Black and White Women Stratified by Estrogen and Progesterone Receptor Statuses (Gleason et al., 2012)	Population-based registry/incidence study	SEER 17	Representative for 2 cells; overlap for 2	PMID 23166647
Recent trends in breast cancer among younger women in the United States (Brinton et al., 2008)	Population-based registry/incidence study	SEER 13	Overlap/sensitivity only (1 cell)	PMID 19001605
Racial and regional disparities of triple negative breast cancer incidence rates in the United States: An analysis of 2011-2019 NPCR and SEER incidence data (Zhang et al., 2022)	Population-based registry/incidence study	NPCR + SEER (USCS)	Overlap/sensitivity only (4 cells)	PMID 36530732
Breast cancer incidence trends in Asian women aged 20 or older as compared to other ethnic women in the United States from 2000 to 2018 by time period, age and tumor stage (Du et al., 2022)	Population-based registry/incidence study	SEER 18	Overlap/sensitivity only (4 cells)	PMID 34861613
Breast Cancer Incidence in Asian American, Native Hawaiian, and Pacific Islander Populations, 2000-2022 (Gomez et al., 2026)	Population-based registry/incidence study	SEER-21	Representative for 11 cells; overlap for 5	PMID 42377954
Hidden breast cancer disparities in Asian women: disaggregating incidence rates by ethnicity and migrant status. (Gomez et al., 2010)	Population-based registry/incidence study	California Cancer Registry	Overlap/sensitivity only (6 cells)	PMID 20147696
Cancer incidence in indigenous people in Australia, New Zealand, Canada, and the USA: A comparative population-based study (2015)	Population-based registry/incidence study	IHS-CHSDA	Overlap/sensitivity only (1 cell)	PMID 26476758
Age-related crossover in breast cancer incidence rates between black and white ethnic groups (Anderson et al., 2008)	Population-based registry/incidence study	SEER 9	Representative for 2 cells	PMID 19066264
Variation in breast cancer subtype incidence and distribution by race/ethnicity in the United States from 2010 to 2015 (Kong et al., 2020)	Population-based registry/incidence study	SEER 18	Representative for 9 cells; overlap for 11	PMID 33074325
Breast cancer in Asian Americans in California, 1988-2013: increasing incidence trends and recent data on breast cancer subtypes (Gomez et al., 2017)	Population-based registry/incidence study	California Cancer Registry	Representative for 2 cells	PMID 28365834
Breast cancer incidence patterns among california hispanic women: Differences by nativity and residence in an enclave (Keegan et al., 2010)	Population-based registry/incidence study	California Cancer Registry	Overlap/sensitivity only (1 cell)	PMID 20447917
Patterns and Trends in Age-Specific Black-White Differences in Breast Cancer Incidence and Mortality - United States, 1999-2014 (Richardson et al., 2016)	Population-based registry/incidence study	USCS	Overlap/sensitivity only (1 cell)	PMID 27736827
Differences in breast cancer incidence among young women aged 20-49 years by stage and tumor characteristics, age, race, and ethnicity, 2004-2013 (Shoemaker et al., 2018)	Population-based registry/incidence study	NPCR + SEER	Representative for 4 cells	PMID 29445940
Age/race differences in HER2 testing and in incidence rates for breast cancer triple subtypes: A population-based study and first report (Lund et al., 2010)	Population-based registry/incidence study	Atlanta SEER + Georgia registry	Overlap/sensitivity only (2 cells)	PMID 20336785
Higher population-based incidence rates of triple-negative breast cancer among young African-American women (Amirikia et al., 2011)	Population-based registry/incidence study	California Cancer Registry	Overlap/sensitivity only (2 cells)	PMID 21656753
The changing incidence of in situ and invasive ductal and lobular breast carcinomas: United States, 1999-2004 (2009)	Population-based registry/incidence study	NPCR+SEER	Overlap/sensitivity only (3 cells)	PMID 19454615

Study (author, year)	Study design	Data source	Role in synthesis	PMID / DOI
Trends and Patterns of Late and Unstaged Lung, Colorectal, Female Breast, and Prostate Cancers among American Indians in the Northern Plains, 2002-2009 (Watanabe-Galloway et al., 2015)	Population-based registry/incidence study	NE/ND/SD state registries	Overlap/sensitivity only (1 cell)	PMID 26320932
Trends in area-socioeconomic and race-ethnic disparities in breast cancer incidence, stage at diagnosis, screening, mortality, and survival among women ages 50 years and over (1987-2005) (Harper et al., 2009)	Population-based registry/incidence study	SEER 9	Representative for 4 cells	PMID 19124489
Trends and variations in breast and colorectal cancer incidence from 1995 to 2011: A comparative study between Texas Cancer Registry and National Cancer Institute's Surveillance, Epidemiology and End Results data (2015)	Population-based registry/incidence study	Texas (state)	Overlap/sensitivity only (1 cell)	PMID 25672365
Trends in Cancer Incidence among American Indians and Alaska Natives and Non-Hispanic Whites in the United States, 1999-2015 (Gopalani et al., 2020)	Population-based registry/incidence study	CDC USCS (NPCR+SEER)	Overlap/sensitivity only (1 cell)	PMID 31764279
Characterizing breast cancer incidence and trends among Asian American, Native Hawaiian, and non-Hispanic White women in Hawai'i, 1990-2014 (Ihenacho et al., 2023)	Population-based registry/incidence study	Hawaii Tumor Registry	Representative for 6 cells	PMID 36504334
Breast cancer in the middle eastern population of California, 1988-2004 (Nasseri, 2009)	Population-based registry/incidence study	California Cancer Registry	Representative for 1 cell	PMID 19292805
Cancer among Asian Indians/Pakistanis living in the United States: Low incidence and generally above average survival (Goggins et al., 2009)	Population-based registry/incidence study	SEER	Overlap/sensitivity only (1 cell)	PMID 19067192
Cancer incidence, stage at diagnosis, and trends across the Navajo Nation, 2014-2018 (Yazzie et al., 2025)	Population-based registry/incidence study	NM Tumor Registry + AZ + UT (IHS-linked)	Representative for 1 cell	PMID 41385397
Associations of Obesity, Physical Activity, and Screening With State-Level Trends and Racial and Ethnic Disparities of Breast Cancer Incidence and Mortality in the US (Xie et al., 2022)	Population-based registry/incidence study	USCS	Overlap/sensitivity only (4 cells)	PMID 35699957
Cancer disparities among non-Hispanic urban American Indian and Alaska Native populations in the United States, 1999-2017 (Melkonian et al., 2022)	Population-based registry/incidence study	NPCR+SEER (IHS-linked)	Overlap/sensitivity only (1 cell)	PMID 35119703
Breast Cancer Incidence, Hormone Receptor Status, Historical Redlining, and Current Neighborhood Characteristics in Massachusetts, 2005-2015 (2022)	Population-based registry/incidence study	Massachusetts Cancer Registry	Overlap/sensitivity only (4 cells)	PMID 35603845
Subtype-specific breast cancer incidence rates in black versus white men in the United States (Sung et al., 2020)	Population-based registry/incidence study	NPCR+SEER	Representative for 3 cells; overlap for 2	PMID 32337499
Disparities in cancer incidence and trends among American Indians and Alaska natives in the United States, 2010-2015 (Melkonian et al., 2019)	Population-based registry/incidence study	NPCR (IHS-linked)	Representative for 5 cells; overlap for 2	PMID 31575554
Cancer incidence and multilevel measures of residential economic and racial segregation for cancer registries (2018)	Population-based registry/incidence study	Massachusetts Cancer Registry	Overlap/sensitivity only (4 cells)	PMID 31360840
Cancer incidence in first generation U.S. Hispanics: Cubans, Mexicans, Puerto Ricans, and New Latinos (Pinheiro et al., 2009)	Population-based registry/incidence study	Florida Cancer Data System	Representative for 4 cells	PMID 19661072
Cambodian cancer incidence rates in California and	Population-based	SEER (CA + Puget Sound)	Representative for 1 cell	PMID 17654663

Study (author, year)	Study design	Data source	Role in synthesis	PMID / DOI
Washington, 1998-2002 (Kem et al., 2007)	registry/incidence study			
Patterns of cancer incidence among US Hispanics/Latinos, 1995-2000 (Carozza et al., 2006)	Population-based registry/incidence study	NAACCR	Overlap/sensitivity only (1 cell)	PMID 16933057
Cancer incidence patterns in Koreans in the US and in Kangwha, South Korea (Gomez et al., 2003)	Population-based registry/incidence study	SEER + California	Overlap/sensitivity only (1 cell)	PMID 12749722
Incidence of and Trends in the Leading Cancers with Elevated Incidence among American Indian and Alaska Native Populations, 2012-2016 (Melkonian et al., 2021)	Population-based registry/incidence study	NPCR+SEER (IHS-linked)	Representative for 1 cell; overlap for 1	doi:10.1093/aje/kwaa222
Neighborhood social determinants of triple negative breast cancer (2019)	Population-based registry/incidence study	Louisiana Tumor Registry	Overlap/sensitivity only (1 cell)	PMID 30834239
Cancer incidence among Asian American populations in the United States, 2009-2011 (Jin et al., 2016)	Population-based registry/incidence study	SEER + state (NPCR)	Overlap/sensitivity only (7 cells)	PMID 26661680
Cancer incidence and mortality patterns among specific Asian and Pacific Islander populations in the US (Miller et al., 2008)	Population-based registry/incidence study	SEER + state	Overlap/sensitivity only (1 cell)	PMID 18066673
Cancer incidence, mortality, and associated risk factors among Asian Americans of Chinese, Filipino, Vietnamese, Korean, and Japanese ethnicities (McCracken et al., 2007)	Population-based registry/incidence study	California Cancer Registry	Overlap/sensitivity only (6 cells)	PMID 17626117
Cancer among Hispanic women in South Florida: An 18-year assessment - A report from the Florida Cancer Data System (Wilkinson et al., 2002)	Population-based registry/incidence study	Florida Cancer Data System	Overlap/sensitivity only (1 cell)	PMID 12365024
Narrative synthesis only (n = 118)				
Theme: Nativity & immigrant generation (n = 8)				
Breast, Colorectal, and Prostate Cancer Incidence among Filipino Americans by Generational Status in the Multiethnic Cohort Study (Abe et al., 2024)	Cohort study	Multiethnic Cohort	No NHW-referenced rate or ratio could be recovered	PMID 39132985
The role of ethnic enclaves and neighborhood socioeconomic status in invasive breast cancer incidence rates among Asian American, Native Hawaiian, and Pacific Islander females in California (Truong et al., 2025)	Population-based registry/incidence study	California Cancer Registry	No NHW-referenced rate or ratio could be recovered	PMID 39438356
Cancer health disparities among Indian and Pakistani immigrants in the United States a Surveillance, Epidemiology, and End Results-based study from 1988 to 2003 (Hossain et al., 2008)	Population-based registry/incidence study	SEER 17	Reports a proportional incidence measure, not an incidence rate	PMID 18696716
Cancer incidence among Korean-American immigrants in the United States and native Koreans in South Korea (Lee et al., 2007)	Population-based registry/incidence study	SEER	The comparison group is a non-U.S. population	PMID 17242674
Cancer in the Minnesota Hmong population (Ross et al., 2003)	Population-based registry/incidence study	Minnesota Cancer Surveillance System	Reports a proportional incidence measure, not an incidence rate	PMID 12784344
Cancer Incidence Profile in Sub-Saharan African-Born Blacks in the United States: Similarities and Differences With US-Born Non-Hispanic Blacks (Medhanie et al., 2017)	Population-based registry/incidence study	SEER 17	Reports a proportional incidence measure, not an incidence rate	PMID 28407201
Differences in the cancer burden among foreign-born and US-born Arab Americans living in metropolitan Detroit (Khan et al., 2013)	Population-based registry/incidence study	Metropolitan Detroit (MDCSS)	Reports a proportional incidence measure, not an incidence rate	PMID 24013772
Cancer risk in different generations of Middle Eastern immigrants to California, 1988-2013 (Ziadeh et al., 2017)	Population-based registry/incidence study	California Cancer Registry	Reports a proportional incidence measure, not an incidence rate	PMID 28801942

Study (author, year)	Study design	Data source	Role in synthesis	PMID / DOI
Theme: Molecular subtype & histology (n = 28)				
Incidence Rate Trends of Breast Cancer Overall and by Molecular Subtype by Race and Ethnicity and Age (Li et al., 2025)	Population-based registry/incidence study	SEER 22	Reports an annual trend, not a cross-sectional rate or ratio	PMID 39853979
Incidence and survival of inflammatory breast cancer between 1973 and 2015 in the SEER database (2021)	Population-based registry/incidence study	SEER	No NHW-referenced rate or ratio could be recovered	PMID 33033965
Incidence of inflammatory breast cancer in women, 1992-2009, United States (2014)	Population-based registry/incidence study	SEER	No NHW-referenced rate or ratio could be recovered	PMID 24366421
Age-specific incidence of breast cancer subtypes: Understanding the black-white crossover (Clarke et al., 2012)	Population-based registry/incidence study	California Cancer Registry	No NHW-referenced rate or ratio could be recovered	PMID 22773826
Incidence trends of breast cancer molecular subtypes by age and race/ethnicity in the US from 2010 to 2016 (Acheampong et al., 2020)	Population-based registry/incidence study	SEER 18	Reports an annual trend, not a cross-sectional rate or ratio	PMID 32804214
Decreasing Incidence of Estrogen Receptor-Negative Breast Cancer in the United States: Trends by Race and Region (Davis Lynn et al., 2022)	Population-based registry/incidence study	SEER 12/17	Reports an annual trend, not a cross-sectional rate or ratio	PMID 34508608
Are There Regional Differences in Triple Negative Breast Cancer among Non-Hispanic Black Women? (Bartley et al., 2021)	Population-based registry/incidence study	USCS	No NHW-referenced rate or ratio could be recovered	PMID 32778443
Breast cancer incidence rates in U.S. women re no longer declining (DeSantis et al., 2011)	Population-based registry/incidence study	SEER 12	Reports an annual trend, not a cross-sectional rate or ratio	PMID 21357727
Estrogen receptor breast cancer phenotypes in the Surveillance, Epidemiology, and End Results database (2002)	Population-based registry/incidence study	SEER	Race-specific rates appear only in a figure, without the values	PMID 12408373
Home mortgage discrimination and incidence of triple-negative and Luminal A breast cancer among non-Hispanic Black and non-Hispanic White females in California, 2006-2015 (Michaels et al., 2022)	Population-based registry/incidence study	California Cancer Registry	No NHW-referenced rate or ratio could be recovered	PMID 35113296
What can we learn from the age- and race/ethnicity- specific rates of inflammatory breast carcinoma? (Il'yasova et al., 2011)	Population-based registry/incidence study	SEER 17	No NHW-referenced rate or ratio could be recovered	PMID 21850396
Adenoid cystic carcinoma of the breast in the United States (1977 to 2006): a population-based cohort study (2010)	Cohort study	SEER	No NHW-referenced rate or ratio could be recovered	PMID 20653964
Recent breast cancer trends among Asian/Pacific Islander, Hispanic, and African-American women in the US: Changes by tumor subtype (Hausauer et al., 2007)	Population-based registry/incidence study	SEER (LA+)	Reports an annual trend, not a cross-sectional rate or ratio	PMID 18162138
Rates for breast cancer characteristics by estrogen and progesterone receptor status in the major racial/ethnic groups (Chu et al., 2002)	Population-based registry/incidence study	SEER 11	No NHW-referenced rate or ratio could be recovered	PMID 12206512
Regional disparities in incidence and outcomes of invasive ductal carcinoma of the breast among Asian and Pacific Islander women in the United States (Bunte et al., 2025)	Population-based registry/incidence study	SEER 17	No NHW-referenced rate or ratio could be recovered	PMID 40543242
Trends in Incidence of Invasive Lobular Carcinoma of the Breast by Race: Patterns by Age, Cancer Stage, and Socioeconomic Factors in the United States, 1992-2019 (Quinn et al., 2025)	Population-based registry/incidence study	SEER 12	Reports an annual trend, not a cross-sectional rate or ratio	PMID 39837694
Changes in breast cancer incidence rates in the	Population-based	SEER	Reports an annual trend, not a cross-sectional	PMID 18086787

Study (author, year)	Study design	Data source	Role in synthesis	PMID / DOI
United States by histologic subtype and race/ethnicity, 1995 to 2004 (Li et al., 2007)	registry/incidence study		rate or ratio	
Racial differences in the incidence of breast cancer subtypes defined by combined histologic grade and hormone receptor status (Cunningham et al., 2010)	Population-based registry/incidence study	South Carolina and Ohio state registries (non-SEER)	No NHW-referenced rate or ratio could be recovered	PMID 20024610
Persistent poverty and breast cancer incidence by tumor subtype: intersections of rural/urban residence and race within USA Surveillance Epidemiology and End Results Registries, 2017 to 2021 (Sherr et al., 2026)	Population-based registry/incidence study	SEER 22	No NHW-referenced rate or ratio could be recovered	PMID 41546752
The greater impact of menopause on ER- than ER+ breast cancer incidence: A possible explanation (United States) (Tarone et al., 2002)	Population-based registry/incidence study	SEER	No NHW-referenced rate or ratio could be recovered	PMID 11899120
Association of inflammatory and noninflammatory breast cancer with socioeconomic characteristics in the surveillance, epidemiology, and end results database, 2000-2007 (2012)	Population-based registry/incidence study	SEER	Compares by socioeconomic position, not against NHW women	PMID 22028401
Trends in breast cancer incidence by estrogen receptor status in the United States, 2004-2020 (2024)	Population-based registry/incidence study	NPCR+SEER	Reports an annual trend, not a cross-sectional rate or ratio	PMID 38985086
Population-based statistics for women diagnosed with inflammatory breast cancer (United States) (Wingo et al., 2004)	Population-based registry/incidence study	26 registries (~40% US)	No NHW-referenced rate or ratio could be recovered	PMID 15090727
Disparities in breast cancer incidence and survival by age, race, and molecular subtype in US women (Wang et al., 2026)	Population-based registry/incidence study	SEER 8/17	No NHW-referenced rate or ratio could be recovered	doi:10.1038/s41523-026-00935-y
Breast Cancer Incidence Trends by Estrogen Receptor Status among Asian American Ethnic Groups, 1990-2014 (Tuan et al., 2021)	Population-based registry/incidence study	SEER 9 + Asian/PI	Reports an annual trend, not a cross-sectional rate or ratio	PMID 33392441
Structural Racism and Triple-Negative Breast Cancer among Black and White Women in the United States (Eldridge et al., 2022)	Population-based registry/incidence study	SEER 18	No NHW-referenced rate or ratio could be recovered	doi:10.1089/heq.2021.0041
Characterizing inflammatory breast cancer among Arab Americans in the California, Detroit and New Jersey Surveillance, Epidemiology and End Results (SEER) registries (1988-2008) (Hirko et al., 2013)	Population-based registry/incidence study	SEER (CA/Detroit/NJ)	No NHW-referenced rate or ratio could be recovered	doi:10.1186/2193-1801-2-3
Asian ethnicity and breast cancer subtypes: A study from the California Cancer Registry (Telli et al., 2011)	Population-based registry/incidence study	California Cancer Registry	No NHW-referenced rate or ratio could be recovered	doi:10.1007/s10549-010-1173-8
Theme: Age & early-onset (n = 20)				
Racial and ethnic disparities in breast cancer rates by age: NAACCR Breast Cancer Project (Joslyn et al., 2005)	Population-based registry/incidence study	NAACCR	No NHW-referenced rate or ratio could be recovered	PMID 15986118
Breast cancer among Asian Indian and Pakistani Americans: A surveillance, epidemiology and end results-based study (Satagopan et al., 2021)	Population-based registry/incidence study	SEER 9 + NJ + San Jose	No NHW-referenced rate or ratio could be recovered	PMID 33099777
Breast cancer racial differences before age 40--implications for screening. (Johnson, 2002)	Population-based registry/incidence study	SEER + 5 state registries	No NHW-referenced rate or ratio could be recovered	PMID 11918384
Rapidly rising breast cancer incidence rates among Asian-American women (Deapen et al., 2002)	Population-based registry/incidence study	LA County / CCR + SEER	Reports an annual trend, not a cross-sectional rate or ratio	PMID 12115511
Trends in breast cancer incidence rates by race/ethnicity: Patterns by stage, socioeconomic position, and geography in the United States, 1999-2017 (Kaur et al., 2022)	Population-based registry/incidence study	NAACCR CiNA	Reports an annual trend, not a cross-sectional rate or ratio	PMID 34731501

Study (author, year)	Study design	Data source	Role in synthesis	PMID / DOI
Trends in stage at diagnosis for young breast cancer patients in the United States (DeSantis et al., 2019)	Population-based registry/incidence study	NAACCR (42 states) + SEER	No NHW-referenced rate or ratio could be recovered	PMID 30426307
Trends in Female Breast Cancer Incidence among Japanese, Korean, and US Populations: An Age-Period-Cohort Analysis (Abubakar et al., 2026)	Population-based registry/incidence study	SEER 8 + Cancer Incidence in Five Continents (CI5)	Reports an annual trend, not a cross-sectional rate or ratio	PMID 42065927
Early- and late-onset breast cancer types among women in the United States and Japan (Matsuno et al., 2007)	Population-based registry/incidence study	SEER 9 + Osaka	The comparison group is a non-U.S. population	PMID 17627009
Racial/ethnicity disparities in invasive breast cancer among younger and older women: An analysis using multiple measures of population health (Hung et al., 2016)	Population-based registry/incidence study	SEER 18 / NPCR-SEER	No NHW-referenced rate or ratio could be recovered	PMID 27792934
Comparison of clinicopathologic features and survival in young American women aged 18-39 years in different ethnic groups with breast cancer (Liu et al., 2013)	Population-based registry/incidence study	SEER 13	No NHW-referenced rate or ratio could be recovered	PMID 23907433
Changes in the pattern of breast cancer burden among African American women: Evidence based on 29 states and District of Columbia during 1998 to 2010 (Sighoko et al., 2015)	Population-based registry/incidence study	SEER + NPCR + NAACCR (29 states)	Reports an annual trend, not a cross-sectional rate or ratio	PMID 25442056
Geographic differences in early-onset breast cancer incidence trends in the USA, 2001-2020, is it time for a geographic risk score? (Kehm et al., 2025)	Population-based registry/incidence study	USCS	Reports an annual trend, not a cross-sectional rate or ratio	PMID 39937364
Cancer statistics for adolescents and young adults, 2020 (Miller et al., 2020)	Population-based registry/incidence study	multi-registry aggregate report	Latest edition of a statistics series; its rates are already supplied by the primary study for that registry and period	PMID 32940362
Early-Onset Cancer Incidence Disparities Between Black and White Individuals in the US, 2003-2022 (Lawrence et al., 2026)	Population-based registry/incidence study	USCS	Reports an annual trend, not a cross-sectional rate or ratio	PMID 41989785
Inequalities in Fertility-Impacting Cancer Incidence Among Young Populations in the United States (Tierney et al., 2025)	Population-based registry/incidence study	SEER	Compares by socioeconomic position, not against NHW women	PMID 40189850
Early-onset cancer incidence in the United States by race/ethnicity between 2011 and 2020 (Gupta et al., 2024)	Population-based registry/incidence study	SEER 22	Reports an annual trend, not a cross-sectional rate or ratio	PMID 39094298
Incidence and survival among young women with stage I-III breast cancer: Seer 2000-2015 (Thomas et al., 2019)	Population-based registry/incidence study	SEER 18	No NHW-referenced rate or ratio could be recovered	PMID 31392297
Cancer in American Indian and Alaska Native Young Adults (Ages 20-44 Years): US, 1999-2004 (Weir et al., 2008)	Population-based registry/incidence study	NPCR+SEER (IHS-linked)	No NHW-referenced rate or ratio could be recovered	PMID 18720386
Incidence of selected cancers in Non-Hispanic American Indian and Alaska Native adolescent and young adult populations, 1999-2019 (Melkonian et al., 2023)	Population-based registry/incidence study	NPCR+SEER (IHS-linked)	No NHW-referenced rate or ratio could be recovered	PMID 37119997
AN EXCESS OF BREAST CANCER AMONG YOUNG CALIFORNIA-BORN ASIAN WOMEN (Reynolds et al., 2011)	Population-based registry/incidence study	California Cancer Registry	No NHW-referenced rate or ratio could be recovered	PMID 21749024
Theme: Socioeconomic status & screening (n = 9)				
Socioeconomic status and breast cancer incidence in California for different race/ethnic groups (Yost et al., 2001)	Population-based registry/incidence study	California Cancer Registry	Compares by socioeconomic position, not against NHW women	PMID 11562110

Study (author, year)	Study design	Data source	Role in synthesis	PMID / DOI
Trends in breast cancer by race and ethnicity: Update 2006 (Smigal et al., 2006)	Population-based registry/incidence study	SEER + NAACCR	Reports an annual trend, not a cross-sectional rate or ratio	PMID 16737949
Does socioeconomic disparity in cancer incidence vary across racial/ethnic groups? (Yin et al., 2010)	Population-based registry/incidence study	California Cancer Registry	Compares by socioeconomic position, not against NHW women	PMID 20567897
Decline in US breast cancer rates after the Women's Health Initiative: socioeconomic and racial/ethnic differentials. (Krieger et al., 2010)	Cohort study	SEER 13	Reports an annual trend, not a cross-sectional rate or ratio	PMID 20147667
Race/ethnicity and changing US socioeconomic gradients in breast cancer incidence: California and Massachusetts, 1978-2002 (United States) (Krieger et al., 2006)	Population-based registry/incidence study	California + Massachusetts (SEER)	Compares by socioeconomic position, not against NHW women	PMID 16425100
Cancer Incidence and Trends in Persistent Poverty Areas of California by Race/Ethnicity and Sex (Movsisyan Vernon et al., 2025)	Population-based registry/incidence study	California Cancer Registry	No NHW-referenced rate or ratio could be recovered	PMID 40736150
The role of area-level socioeconomic disadvantage in racial disparities in cancer incidence in metropolitan Detroit (Purrington et al., 2023)	Population-based registry/incidence study	Metro Detroit (MDCSS)	No NHW-referenced rate or ratio could be recovered	PMID 37184135
Racial and socioeconomic disparities in breast cancer diagnosis and mortality in Pennsylvania (Williams et al., 2022)	Population-based registry/incidence study	Pennsylvania Cancer Registry	No NHW-referenced rate or ratio could be recovered	PMID 35064367
The fluctuating incidence, improved survival of patients with breast cancer, and disparities by age, race, and socioeconomic status by decade, 1981–2010 (Lu et al., 2018)	Population-based registry/incidence study	SEER 9	No NHW-referenced rate or ratio could be recovered	doi:10.2147/cmar.s173099
Theme: Geography & region (n = 24)				
Cumulative Incidence Rates of Breast Cancer by Age for Filipinos and Chamorros in Guam (Badowski et al., 2020)	Population-based registry/incidence study	Guam CR + Hawaii TR + SEER	No NHW-referenced rate or ratio could be recovered	PMID 32596677
Neighborhood composition and incidence of cancer among Hispanics in the United States (2005)	Population-based registry/incidence study	SEER	No NHW-referenced rate or ratio could be recovered	PMID 15672387
Cancer incidence trends among native hawaiians and other pacific islanders in the united states, 1990-2008 (Liu et al., 2013)	Population-based registry/incidence study	SEER	Reports an annual trend, not a cross-sectional rate or ratio	PMID 23878354
Patterns in cancer incidence among American Indians/Alaska Natives, United States, 1992-1999 (Paltoo et al., 2004)	Population-based registry/incidence study	SEER (12 areas, 42% AI/AN)	No NHW-referenced rate or ratio could be recovered	PMID 15219802
Cancer rates among American Indians and Alaska Natives: is there a national perspective (Swan et al., 2003)	Population-based registry/incidence study	SEER + ANTR	No NHW-referenced rate or ratio could be recovered	PMID 12973851
Inter- and intra-ethnic differences for female breast carcinoma incidence in the continental United States and in the state of Hawaii (Fong et al., 2006)	Population-based registry/incidence study	SEER 13	No NHW-referenced rate or ratio could be recovered	PMID 16322891
Cancers of the breast, uterus, ovary and cervix among Alaska native women, 1974-2003 (Day et al., 2010)	Population-based registry/incidence study	Alaska Native Tumor Registry	No NHW-referenced rate or ratio could be recovered	PMID 20167158
Explaining geographic variation in breast and cervical cancer incidence rates in US Hispanic women (Polednak, 2005)	Population-based registry/incidence study	SEER (11 registries)	Race-specific rates appear only in a figure, without the values	PMID 16259500
Disparities in Breast Cancer Incidence, Mortality, and Quality of Care among African American and European American Women in South Carolina (Samson et al., 2016)	Population-based registry/incidence study	State Cancer Profiles (South Carolina)	No NHW-referenced rate or ratio could be recovered	PMID 26741869
Recent trends in breast cancer incidence among 6 Asian groups in the Greater Bay Area of Northern	Population-based registry/incidence study	Greater Bay Area / CCR	Reports an annual trend, not a cross-sectional rate or ratio	PMID 17163416

Study (author, year)	Study design	Data source	Role in synthesis	PMID / DOI
California (Keegan et al., 2007)				
Cancer statistics for Asian Americans, Native Hawaiians, and Pacific Islanders, 2016: Converging incidence in males and females (Torre et al., 2016)	Population-based registry/incidence study	multi-registry aggregate report	Latest edition of a statistics series; its rates are already supplied by the primary study for that registry and period	PMID 26766789
Cancer in Guam and Hawaii: A comparison of two U.S. Island populations (Hernandez et al., 2017)	Population-based registry/incidence study	Guam CR + Hawaii TR	No NHW-referenced rate or ratio could be recovered	PMID 29120826
Secular trends, race, and geographic disparity of early-stage breast cancer incidence: 25 years of surveillance in Connecticut (Crabbe et al., 2015)	Population-based registry/incidence study	Connecticut Tumor Registry	No NHW-referenced rate or ratio could be recovered	PMID 25905822
Urbanization and breast cancer incidence in North Carolina, 1995-1999 (Hall et al., 2005)	Population-based registry/incidence study	North Carolina Central Cancer Registry	No NHW-referenced rate or ratio could be recovered	PMID 15922628
Disparities in cancer incidence by rurality in California (Oh et al., 2023)	Population-based registry/incidence study	California Cancer Registry	No NHW-referenced rate or ratio could be recovered	PMID 36622036
Cancer-related risk factors and incidence of major cancers by race, gender and region; analysis of the NIH-AARP diet and health study (Akinyemiju et al., 2017)	Cohort study	NIH-AARP cohort	No NHW-referenced rate or ratio could be recovered	PMID 28854891
Cancer incidence in the Somali population of Olmsted County: A Rochester epidemiology project study (Mohamed et al., 2023)	Population-based registry/incidence study	Rochester Epidemiology Project	The comparison group is a non-U.S. population	PMID 37740603
Rural-Urban Cancer Incidence and Trends in the United States, 2000 to 2019 (Sempini et al., 2024)	Population-based registry/incidence study	SEER 22	No NHW-referenced rate or ratio could be recovered	PMID 38801414
Cancer incidence and associations with known risk and protective factors: the Alaska EARTH study (Nash et al., 2019)	Population-based registry/incidence study	Alaska EARTH cohort (ANTR)	No NHW-referenced rate or ratio could be recovered	PMID 31428891
Cancer incidence rates among South Asians in four geographic regions: India, Singapore, UK and US (Rastogi et al., 2008)	Population-based registry/incidence study	SEER 11 + state	The comparison group is a non-U.S. population	PMID 18094016
Cancer incidence in the Middle Eastern population of California, 1988-2004. (Nasseri et al., 2007)	Population-based registry/incidence study	California Cancer Registry	No NHW-referenced rate or ratio could be recovered	PMID 18159978
Cancer incidence in the South Asian population of California, 1988-2000 (Jain et al., 2005)	Population-based registry/incidence study	California Cancer Registry	No NHW-referenced rate or ratio could be recovered	doi:10.1186/1477-3163-4-21
The effect of migration on cancer incidence among Japanese in Hawaii (Maskarinec et al., 2004)	Population-based registry/incidence study	Hawaii Tumor Registry	Race-specific rates appear only in a figure, without the values	PMID 15328946
Annual Report to the Nation on the Status of Cancer, featuring state-level statistics after the onset of the COVID-19 pandemic (Sherman et al., 2025)	Population-based registry/incidence study	multi-registry aggregate report	Latest edition of a statistics series; its rates are already supplied by the primary study for that registry and period	PMID 40257373
Theme: Time trends (n = 8)				
Black-white breast cancer incidence trends: Effects of ethnicity (Davis Lynn et al., 2018)	Population-based registry/incidence study	SEER 13	Reports an annual trend, not a cross-sectional rate or ratio	PMID 29982593
Trends in breast cancer by race and ethnicity (Ghafoor et al., 2003)	Population-based registry/incidence study	SEER 9	Reports an annual trend, not a cross-sectional rate or ratio	PMID 15224974
Cancer incidence trends among asian american populations in the united states, 1990-2008 (Gomez et al., 2013)	Population-based registry/incidence study	SEER	Reports an annual trend, not a cross-sectional rate or ratio	PMID 23878350
Forty-year trends in menopausal hormone therapy use and breast cancer incidence among postmenopausal black and white women (Chlebowski et al., 2020)	Population-based registry/incidence study	SEER 9	Reports an annual trend, not a cross-sectional rate or ratio	PMID 32212335
A trend analysis of breast cancer incidence rates in the United States from 2000 to 2009 shows a recent increase (Hou et al., 2013)	Population-based registry/incidence study	SEER 18	Reports an annual trend, not a cross-sectional rate or ratio	PMID 23446808

Study (author, year)	Study design	Data source	Role in synthesis	PMID / DOI
Incidence, Trends, and Racial/Ethnic Disparities in Female Breast Cancer in the United States, 2000-2023: A SEER Based Study (Mzizi et al., 2026)	Population-based registry/incidence study	SEER	No NHW-referenced rate or ratio could be recovered	PMID 42466659
Spillover effects of public school integration in the southern United States: diverging trends in statewide annual breast cancer incidence, 2001-2019 (Chanpura, 2026)	Population-based registry/incidence study	USCS	No NHW-referenced rate or ratio could be recovered	PMID 41649619
Trends in Breast Cancer Incidence and Mortality in the United States From 2004-2018: A Surveillance, Epidemiology, and End Results (SEER)-Based Study (Bazzi et al., 2023)	Population-based registry/incidence study	SEER 18	Reports an annual trend, not a cross-sectional rate or ratio	PMID 37223193
Theme: Other (subgroup-specific descriptive) (n = 21)				
Breast cancer statistics 2024 (Giaquinto et al., 2024)	Population-based registry/incidence study	multi-registry aggregate report	Latest edition of a statistics series; its rates are already supplied by the primary study for that registry and period	PMID 39352042
Cancer statistics for the US Hispanic/Latino population, 2021 (Miller et al., 2021)	Population-based registry/incidence study	multi-registry aggregate report	Latest edition of a statistics series; its rates are already supplied by the primary study for that registry and period	PMID 34545941
Invasive cancer incidence - United States, 2010. (Henley et al., 2014)	Population-based registry/incidence study	multi-registry aggregate report	Latest edition of a statistics series; its rates are already supplied by the primary study for that registry and period	PMID 24670926
Disparities of cancer incidence in Michigan's American Indians: Spotlight on breast cancer (Roen et al., 2014)	Population-based registry/incidence study	Michigan registry (IHS-linked)	No NHW-referenced rate or ratio could be recovered	PMID 24676851
Persistence in breast cancer disparities between African Americans and whites in Wisconsin (Lepeak et al., 2011)	Population-based registry/incidence study	Wisconsin (WCRS)	Reports an annual trend, not a cross-sectional rate or ratio	PMID 21473509
Cancer Incidence Rate Variations among the Chinese, South Asian, and Vietnamese in Massachusetts, 2011-2015 (Knowlton et al., 2020)	Population-based registry/incidence study	Massachusetts Cancer Registry	No NHW-referenced rate or ratio could be recovered	PMID 34128917
Underlying causes of the black-white racial disparity in breast cancer mortality: A population-based analysis (2009)	Population-based registry/incidence study	SEER	Race-specific rates appear only in a figure, without the values	PMID 19584327
Early estimates of cancer incidence for 2015: Expanding to include estimates for white and black races (Lewis et al., 2018)	Population-based registry/incidence study	SEER 18	No NHW-referenced rate or ratio could be recovered	PMID 29509274
Breast cancer incidence and stage at diagnosis in the six US-Affiliated Pacific Islands (Gopalani et al., 2024)	Population-based registry/incidence study	Pacific Regional Central Cancer Registry	No NHW-referenced rate or ratio could be recovered	PMID 38996557
American Cancer Society's Report on the Status of Cancer Disparities in the United States, 2025 (Islami et al., 2026)	Population-based registry/incidence study	multi-registry aggregate report	Latest edition of a statistics series; its rates are already supplied by the primary study for that registry and period	PMID 41400551
Cancer statistics, 2026 (Siegel et al., 2026)	Population-based registry/incidence study	multi-registry aggregate report	Latest edition of a statistics series; its rates are already supplied by the primary study for that registry and period	PMID 41528114
Cancer Incidence in the Hispanic Community Health Study/Study of Latinos (HCHS/SOL)—The Onco-SOL Ancillary Study (Parada et al., 2025)	Population-based registry/incidence study	Hispanic Community Health Study/Study of Latinos (HCHS/SOL) + state registries	No NHW-referenced rate or ratio could be recovered	PMID 39808161
Cancer patterns in Hmong in Minnesota, 2000 to 2012 (Oyenuga et al., 2018)	Population-based registry/incidence study	Minnesota Cancer Surveillance System	No NHW-referenced rate or ratio could be recovered	PMID 29975407
Incidence rate for breast cancer in Japanese in Japan and in the United States from the Cancer Incidence in Five Continents (Katanoda et al., 2016)	Population-based registry/incidence study	Cancer Incidence in Five Continents (CI5) + SEER 9/Hawaii	No NHW-referenced rate or ratio could be recovered	doi:10.1093/jjco/hyw119

Study (author, year)	Study design	Data source	Role in synthesis	PMID / DOI
Fifty years of cancer in an American Indian population (Mahoney et al., 2009)	Population-based registry/incidence study	New York State Cancer Registry	No NHW-referenced rate or ratio could be recovered	PMID 19109819
Risk of cancer in Asian Americans: a Kaiser Permanente cohort study (Tran et al., 2016)	Cohort study	Kaiser Permanente (cohort)	No NHW-referenced rate or ratio could be recovered	doi:10.1007/s10552-016-0798-2
Changes in cancer incidence patterns among a northeastern American Indian population: 1955-1969 versus 1990-2004 (Mahoney et al., 2009)	Population-based registry/incidence study	New York State Cancer Registry (IHS-linked)	No NHW-referenced rate or ratio could be recovered	doi:10.1111/j.1748-0361.2009.00247.x
Cancer incidence patterns among Vietnamese in the United States and Ha Noi, Vietnam (Le et al., 2002)	Population-based registry/incidence study	California Cancer Registry + SEER	The comparison group is a non-U.S. population	doi:10.1002/ijc.10725
Racial/ethnic patterns in lifetime and age-conditional risk estimates for selected cancers (Miller et al., 2006)	Population-based registry/incidence study	SEER 9	Compares by socioeconomic position, not against NHW women	PMID 16388524
White-Black Differences in Cancer Incidence, Stage at Diagnosis, and Survival Among Older Adults (Krok-Schoen et al., 2018)	Population-based registry/incidence study	SEER 18	Reports an annual trend, not a cross-sectional rate or ratio	PMID 28553811
Cancer statistics for African American and Black people, 2025 (Saka et al., 2025)	Population-based registry/incidence study	multi-registry aggregate report	Latest edition of a statistics series; its rates are already supplied by the primary study for that registry and period	PMID 39976243

Note. All 176 studies are included in the systematic review; not all entered the quantitative synthesis, and rows are grouped by contribution: 58 were eligible for quantitative synthesis and all provided extractable quantitative data that entered the analysis, and the remaining 118 contributed to the narrative synthesis only (176 = 58 + 118). Study design is classified from the data source; most are population-based registry/incidence studies rather than cohort studies. A PMID (or DOI where none exists) is given for every study so each can be located individually.

Role in synthesis summarises, for each study, how it was used: “Representative for N cell(s)” — supplied the main-analysis benchmark for N analytic cells (group × dimension); “overlap for M” — also contributed M overlapping estimates kept only for the sensitivity re-selection; “Overlap/sensitivity only” — every estimate overlapped an already-represented cell. For the studies under “Narrative synthesis only” the column instead gives the reason no quantitative estimate was taken: the specific reason where the eligibility record states one (a recurring statistics series, an annual-trend-only report, a proportional incidence measure, values shown only in a figure, a socioeconomic rather than racial comparison, or a non-U.S. comparison group), otherwise the general reason that no NHW-referenced rate or ratio could be recovered. The cell-by-cell selection status and the reason each estimate was or was not chosen as the representative are given in Supplementary Table 4 (main-analysis column).

Supplementary Table 3. Full-text exclusions with reasons

Study (author, year)	Exclusion reason
Stage IV Breast Cancer Incidence and Survival, 2010-2021 (Avila et al., 2026)	Did not report the outcome of interest
Breast cancer among hispanic and non-hispanic white women in Arizona (Martínez et al., 2007)	Did not report the outcome of interest
Triple-negative breast cancer incidence in the United States: ecological correlations with area-level sociodemographics, healthcare, and health behaviors (Moss et al., 2021)	Did not report the outcome of interest
Race/Ethnicity and Age Distribution of Breast Cancer Diagnosis in the United States (Stapleton et al., 2018)	Did not report the outcome of interest
Incidence of breast cancer with distant involvement among women in the United States, 1976 to 2009 (Johnson et al., 2013)	Did not report the outcome of interest
Risks of developing breast and colorectal cancer in association with incomes and geographic locations in Texas: A retrospective cohort study (Liu et al., 2016)	Did not report the outcome of interest
Disparities in Stage at Presentation for Disaggregated Asian American, Native Hawaiian, and Pacific Islander Patients with Breast Cancer (Thaploo et al., 2025)	Did not report the outcome of interest
The risk of developing breast cancer within the next 5, 10, or 20 years of a woman's life (Morris et al., 2001)	Did not report the outcome of interest
Breast cancer trends among young women in the United States (Tarone, 2006)	Did not report the outcome of interest
Breast Cancer Incidence in Asian American, Native Hawaiian, and Pacific Islander Populations (Calip et al., 2026)	Did not report the outcome of interest
Proportionate cancer incidence in the Laotian population of California, 1988-2006 (2009)	Did not report the outcome of interest
Is male breast cancer similar or different than female breast cancer? (Anderson et al., 2004)	Did not report the outcome of interest
The Impact of Eliminating Socioeconomic and Racial Disparities on Premature Cancer Deaths (Siegel et al., 2011)	Did not report the outcome of interest
Breast cancer trends and outcomes among young Black Women: A national cancer database analysis (Thompson et al., 2026)	Ineligible population
Rising Breast Cancer Incidence and Poor Outcomes in Young Women: A Retrospective Study (Den et al., 2026)	Ineligible population
Differences in Breast Cancer Subtypes among Racial/Ethnic Groups (Sasaki et al., 2024)	Ineligible population
Cancer incidence in the U.S. military population: Comparison with rates from the SEER Program (2009)	Ineligible population
Characterizing cancer patterns in Okinawan vs. mainland Japanese Americans: The Multiethnic Cohort Study (Streicher et al., 2025)	Preprint (not peer-reviewed)
Breast cancer statistics, 2019 (DeSantis et al., 2019)	Duplicate or redundant dataset
Lobular breast cancer statistics, 2025 (Giaquinto et al., 2025)	Duplicate or redundant dataset
Breast cancer statistics, 2015: Convergence of incidence rates between black and white women (DeSantis et al., 2016)	Duplicate or redundant dataset
Cancer statistics for African Americans, 2016: Progress and opportunities in reducing racial disparities (DeSantis et al., 2016)	Duplicate or redundant dataset
Annual report to the nation on the status of cancer, part 1: National cancer statistics (Cronin et al., 2022)	Duplicate or redundant dataset
Annual report to the nation on the status of cancer, part I: National cancer statistics (Henley et al., 2020)	Duplicate or redundant dataset
Annual Report to the Nation on the status of cancer, 1975-2010, featuring prevalence of comorbidity and impact on survival among persons with lung, colorectal, breast, or prostate cancer (Edwards et al., 2014)	Duplicate or redundant dataset
Annual Report to the Nation on the Status of Cancer, 1975-2014, Featuring Survival (Jemal et al., 2017)	Duplicate or redundant dataset
Annual report to the nation on the status of cancer, 1975-2003, featuring cancer among U.S. Hispanic/Latino populations (Howe et al., 2006)	Duplicate or redundant dataset
Annual report to the nation on the status of cancer, 1975-2011, featuring incidence of breast cancer subtypes by race/ethnicity, poverty, and state (Kohler et al., 2015)	Duplicate or redundant dataset
Annual report to the nation on the status of cancer, 1975-2002, featuring population-based trends in cancer treatment (Edwards et al., 2005)	Duplicate or redundant dataset
Breast cancer statistics, 2017, racial disparity in mortality by state (DeSantis et al., 2017)	Duplicate or redundant dataset
Cancer statistics for African American/Black People 2022 (Giaquinto et al., 2022)	Duplicate or redundant dataset
Annual report to the nation on the status of cancer, 1973-1999, featuring implications of age and aging on U.S. cancer burden (Edwards et al., 2002)	Duplicate or redundant dataset
Breast Cancer Statistics, 2022 (Giaquinto et al., 2022)	Duplicate or redundant dataset
Annual report to the nation on the status of cancer, 1975-2004, featuring cancer in American Indians and Alaska	Duplicate or redundant dataset

Study (author, year)	Exclusion reason
Natives (Espey et al., 2007)	
Annual report to the nation on the status of cancer (1973 through 1998), featuring cancers with recent increasing trends (Howe et al., 2001)	Duplicate or redundant dataset
Annual Report to the Nation on the Status of Cancer, Featuring Cancer in Men and Women Age 20–49 Years (Ward et al., 2019)	Duplicate or redundant dataset
The annual report to the nation on the status of cancer, 1973-1997, with a special section on colorectal cancer (Ries et al., 2000)	Duplicate or redundant dataset
Annual report to the nation on the status of cancer, 1975-2006, featuring colorectal cancer trends and impact of interventions (risk factors, screening, and treatment) to reduce future rates (Edwards et al., 2010)	Duplicate or redundant dataset
Cancer statistics, 2012 (Siegel et al., 2012)	Duplicate or redundant dataset
Regional Disparities in Incidence and Outcomes of Invasive Ductal Carcinoma Among Asian and Pacific Islander Women in the United States (Bunte et al., 2024)	Duplicate or redundant dataset
Cancer statistics, 2024 (Siegel et al., 2024)	Duplicate or redundant dataset
Cancer statistics, 2023 (Siegel et al., 2023)	Duplicate or redundant dataset
American Cancer Society's report on the status of cancer disparities in the United States, 2021 (Islami et al., 2022)	Duplicate or redundant dataset
Estrogen receptor negative breast cancer incidence rates are similar in Ghanaian and Non-Hispanic Black women in the USA (Davis Lynn et al., 2022)	Duplicate or redundant dataset
Cancer statistics, 2022 (Siegel et al., 2022)	Duplicate or redundant dataset
Cancer statistics, 2020 (Siegel et al., 2020)	Duplicate or redundant dataset
Cancer statistics for African Americans, 2019 (DeSantis et al., 2019)	Duplicate or redundant dataset
Cancer Statistics for Hispanics/Latinos, 2018 (Miller et al., 2018)	Duplicate or redundant dataset
Cancer statistics, 2018 (Siegel et al., 2018)	Duplicate or redundant dataset
Cancer Statistics, 2017 (Siegel et al., 2017)	Duplicate or redundant dataset
Cancer statistics for Hispanics/Latinos, 2015 (Siegel et al., 2015)	Duplicate or redundant dataset
Cancer statistics, 2015 (Siegel et al., 2015)	Duplicate or redundant dataset
Breast cancer statistics, 2013 (Desantis et al., 2014)	Duplicate or redundant dataset
Cancer statistics for African Americans, 2013 (Desantis et al., 2013)	Duplicate or redundant dataset
Age-Related Crossover in Breast Cancer Incidence Rates between Black and White Women: Implications for Removing Racial Disparities in Breast Cancer Care (Anderson et al., 2009)	Duplicate or redundant dataset
Cancer statistics, 2009 (Jemal et al., 2009)	Duplicate or redundant dataset
Cancer statistics, 2008 (Jemal et al., 2008)	Duplicate or redundant dataset
Cancer statistics, 2005 (Jemal et al., 2005)	Duplicate or redundant dataset
Cancer Statistics, 2004 (Jemal et al., 2004)	Duplicate or redundant dataset
Cancer statistics, 2002 (Jemal et al., 2002)	Duplicate or redundant dataset
Annual Report to the Nation on the Status of Cancer, part 2: Early assessment of the COVID-19 pandemic's impact on cancer diagnosis (Negoita et al., 2024)	Duplicate or redundant dataset
Cancer incidence and mortality among Filipinos in the USA and the Philippines: Patterns and trends (Gomez et al., 2016)	Duplicate or redundant dataset
Annual report to the nation on the status of cancer, 1975-2001, with a special feature regarding survival (Jemal et al., 2004)	Duplicate or redundant dataset
Annual Report to the Nation on the Status of Cancer, 1975-2009, Featuring the Burden and Trends in Human Papillomavirus (HPV)-Associated Cancers and HPV Vaccination Coverage Levels (Jemal et al., 2013)	Duplicate or redundant dataset
Breast cancer statistics, 2011 (DeSantis et al., 2011)	Duplicate or redundant dataset
Cancer Statistics, 2014 (Siegel et al., 2014)	Duplicate or redundant dataset
Annual Report to the Nation on the Status of Cancer, 1975-2007, Featuring Tumors of the Brain and Other Nervous System (Kohler et al., 2011)	Duplicate or redundant dataset
Annual report to the nation on the status of cancer, 1975-2000, featuring the uses of surveillance data for cancer prevention and control (Weir et al., 2003)	Duplicate or redundant dataset

Study (author, year)	Exclusion reason
Annual Report to the Nation on the Status of Cancer, 1975-2005, Featuring Trends in Lung Cancer, Tobacco Use, and Tobacco Control (Jemal et al., 2008)	Duplicate or redundant dataset
Cancer statistics, 2013 (Siegel et al., 2013)	Duplicate or redundant dataset
Cancer statistics for Hispanics/Latinos, 2012 (Siegel et al., 2012)	Duplicate or redundant dataset
Annual Report to the Nation on the status of cancer, 1975-2008, featuring cancers associated with excess weight and lack of sufficient physical activity (Eheman et al., 2012)	Duplicate or redundant dataset
Cancer statistics for Hispanics, 2003 (O'Brien et al., 2003)	Duplicate or redundant dataset
Breast cancer trends among black and white women in the United States (Jatoi et al., 2005)	Full text unavailable
Breast cancer patterns and lifetime risk of developing breast cancer among Puerto Rican females. (Nazario et al., 2000)	Full text unavailable
Ethnic Enclave Residence and Breast, Cervical, and Colorectal Cancer Incidence among Asian American Adults, 2006 to 2017 (2026)	Full text unavailable
Women's cancers among Alaska Natives 1969-2003. (Day et al., 2006)	Full text unavailable
Breast cancer in a multiethnic cohort in Hawaii and Los Angeles: Risk factor-adjusted incidence in Japanese equals and in Hawaiians exceeds that in whites (Pike et al., 2002)	Full text unavailable
Women's cancers among Alaska Natives: [corrected] 1969-2003. (Kelly et al., 2007)	Full text unavailable
Rural-urban disparities in breast cancer incidence among US women aged 20-49: trends by race/ethnicity, stage, poverty, and state from 2000-2019 (Ho et al., 2026)	Full text unavailable
Incidence trends in triple-negative breast cancer among women in the United States from 2010 to 2019 by race/ethnicity, age and tumor stage (Du et al., 2023)	Full text unavailable
Time trends and racial differences in female breast cancer incidence in Pennsylvania, 1985-2004 (Han et al., 2011)	Full text unavailable
Improving cancer incidence estimates for American Indians in Wisconsin (Foote et al., 2007)	Full text unavailable

Supplementary Table 4. Registry overlap and representative selection

Study	Dimension	Group	Comparator	Registry family	Period	Std pop	IRR [95% CI]	Main analysis
Zahrieh 2021 [20]	AI/AN by region	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	State: New Mexico (SEER)	2005-2014	not stated	0.384 [0.253, 0.546]	no (AI/AN undercount)
Wingo 2008 [21]	AI/AN by region	AI/AN (East)	non-Hispanic White (NHW)	IHS-PRCDA	1999-2004	2000 US	0.53 [0.45, 0.63]	no (overlaps representative)
Melkonian 2019 [53]	AI/AN by region	AI/AN (East)	non-Hispanic White (NHW)	IHS-PRCDA	2010-2015	2000 US	0.61	yes (representative)
Yazzie 2025 [48]	AI/AN by region	AI/AN (Navajo)	non-Hispanic White (NHW)	IHS-PRCDA	2014-2018	2000 US	0.49 [0.44, 0.55]	yes (representative)
Wingo 2008 [21]	AI/AN by region	AI/AN (Northern Plains)	non-Hispanic White (NHW)	IHS-PRCDA	1999-2004	2000 US	0.89 [0.81, 0.97]	no (overlaps representative)
Melkonian 2019 [53]	AI/AN by region	AI/AN (Northern Plains)	non-Hispanic White (NHW)	IHS-PRCDA	2010-2015	2000 US	1.05	yes (representative)
Watanabe-Galloway 2015 [41]	AI/AN by region	AI/AN (Northern Plains)	non-Hispanic White (NHW)	Multi-state registry subset (regional)	2002-2009	2000 US	0.902	no (AI/AN undercount)
Wingo 2008 [21]	AI/AN by region	AI/AN (Pacific Coast)	non-Hispanic White (NHW)	IHS-PRCDA	1999-2004	2000 US	0.52 [0.48, 0.57]	no (overlaps representative)
Melkonian 2019 [53]	AI/AN by region	AI/AN (Pacific Coast)	non-Hispanic White (NHW)	IHS-PRCDA	2010-2015	2000 US	0.93	yes (representative)
Wingo 2008 [21]	AI/AN by region	AI/AN (Southern Plains)	non-Hispanic White (NHW)	IHS-PRCDA	1999-2004	2000 US	0.89 [0.83, 0.96]	no (overlaps representative)
Melkonian 2019 [53]	AI/AN by region	AI/AN (Southern Plains)	non-Hispanic White (NHW)	IHS-PRCDA	2010-2015	2000 US	1.30	no (overlaps representative)
Melkonian 2021 [58]	AI/AN by region	AI/AN (Southern Plains)	non-Hispanic White (NHW)	IHS-PRCDA	2012-2016	2000 US	1.33 [1.26, 1.41]	yes (representative)
Wingo 2008 [21]	AI/AN by region	AI/AN (Southwest)	non-Hispanic White (NHW)	IHS-PRCDA	1999-2004	2000 US	0.41 [0.37, 0.44]	no (overlaps representative)
Melkonian 2019 [53]	AI/AN by region	AI/AN (Southwest)	non-Hispanic White (NHW)	IHS-PRCDA	2010-2015	2000 US	0.57	yes (representative)
Nash 2019 [11]	AI/AN by region	Alaska Native	White (not NH-stratified)	Alaska Native Tumor Registry (IHS-linked)	2009-2014	2000 US	1.09 [0.99, 1.21]	yes (representative)
Nash 2022 [12]	AI/AN by region	Alaska Native	non-Hispanic White (NHW)	Alaska Native Tumor Registry (IHS-linked)	2014-2018	2000 US	0.952 [0.851, 1.065]	no (external comparator)
Wingo 2008 [21]	AI/AN by region	Alaska Native	non-Hispanic White (NHW)	IHS-PRCDA	1999-2004	2000 US	0.99 [0.87, 1.12]	no (overlaps representative)
Melkonian 2019 [53]	AI/AN by region	Alaska Native	non-Hispanic White (NHW)	IHS-PRCDA	2010-2015	2000 US	1.26	no (overlaps representative)
Moore 2015 [32]	AI/AN by region	Alaska Native	White (not NH-stratified)	IHS-PRCDA	2002-2006	world standard (IARC)	1.14 [1.01, 1.30] (SIR)	no (overlaps representative)
Melkonian 2021 [58]	AI/AN by region	Alaska Native	non-Hispanic White (NHW)	IHS-PRCDA	2012-2016	2000 US	1.25 [1.11, 1.41]	no (overlaps representative)
Pinheiro 2009 [8]	Hispanic by origin	Cuban	non-Hispanic White (NHW)	State: Florida	1999-2001	2000 US	0.769 [0.688, 0.860]	yes (representative)
Pinheiro 2009 [8]	Hispanic by origin	Mexican	non-Hispanic White (NHW)	State: Florida	1999-2001	2000 US	0.512 [0.382, 0.686]	yes (representative)
Pinheiro 2009 [8]	Hispanic by origin	New Latino	non-Hispanic White (NHW)	State: Florida	1999-2001	2000 US	0.697 [0.629, 0.771]	yes (representative)

Study	Dimension	Group	Comparator	Registry family	Period	Std pop	IRR [95% CI]	Main analysis
Pinheiro 2009 [8]	Hispanic by origin	Puerto Rican	non-Hispanic White (NHW)	State: Florida	1999-2001	2000 US	0.833 [0.738, 0.939]	yes (representative)
Anderson 2008 [33]	Age ≥40	non-Hispanic Black (NHB)	White (not NH-stratified)	SEER-national	1975-2004	2000 US	0.851 [0.841, 0.862]	yes (representative)
Harper 2009 [42]	Age ≥50	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	SEER-national	1997-2001	2000 US	0.647	yes (representative)
Harper 2009 [42]	Age ≥50	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	SEER-national	1997-2001	2000 US	0.659	yes (representative)
Harper 2009 [42]	Age ≥50	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER-national	1997-2001	2000 US	0.914	yes (representative)
Ihenacho 2023 [45]	Age ≥50	Filipina	non-Hispanic White (NHW)	Hawaii Tumor Registry (SEER)	2010-2014	2000 US	0.774 [0.700, 0.855]	yes (representative)
Harper 2009 [42]	Age ≥50	Hispanic	non-Hispanic White (NHW)	SEER-national	1997-2001	2000 US	0.660	yes (representative)
Ihenacho 2023 [45]	Age ≥50	Japanese	non-Hispanic White (NHW)	Hawaii Tumor Registry (SEER)	2010-2014	2000 US	1.064 [0.978, 1.157]	yes (representative)
Ihenacho 2023 [45]	Age ≥50	Native Hawaiian	non-Hispanic White (NHW)	Hawaii Tumor Registry (SEER)	2010-2014	2000 US	1.366 [1.246, 1.499]	yes (representative)
LeeArgov 2024 [17]	age-ge65	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2001-2019	2000 US	0.72 [0.70, 0.73]	yes (representative)
LeeArgov 2024 [17]	age-ge65	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2001-2019	2000 US	0.62 [0.62, 0.63]	yes (representative)
LeeArgov 2024 [17]	age-ge65	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2001-2019	2000 US	0.93 [0.93, 0.94]	yes (representative)
LeeArgov 2024 [17]	age-ge65	Hispanic	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2001-2019	2000 US	0.70 [0.69, 0.70]	yes (representative)
Baquet 2008 [27]	Age <40	non-Hispanic Black (NHB)	White (not NH-stratified)	SEER-national	1995-2004	2000 US	1.16 [1.1, 1.23]	no (overlaps representative)
Anderson 2008 [33]	Age <40	non-Hispanic Black (NHB)	White (not NH-stratified)	SEER-national	1975-2004	2000 US	1.183 [1.142, 1.226]	yes (representative)
Shoemaker 2018 [37]	Age <50	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2004-2013	2000 US	0.70 [0.68, 0.73]	yes (representative)
Xu 2024 [22]	Age <50	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	SEER-national	2000-2019	2000 US	0.75 [0.72, 0.79]	no (overlaps representative)
Shoemaker 2018 [37]	Age <50	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2004-2013	2000 US	0.85 [0.84, 0.86]	yes (representative)
Xu 2024 [22]	Age <50	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	SEER-national	2000-2019	2000 US	0.96 [0.95, 0.97]	no (overlaps representative)
Shoemaker 2018 [37]	Age <50	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2004-2013	2000 US	1.03 [1.02, 1.04]	yes (representative)
Xu 2024 [22]	Age <50	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER-national	2000-2019	2000 US	1.01 [1.0, 1.03]	no (overlaps representative)
Ihenacho 2023 [45]	Age <50	Filipina	non-Hispanic White (NHW)	Hawaii Tumor Registry (SEER)	2010-2014	2000 US	0.768 [0.633, 0.932]	yes (representative)

Study	Dimension	Group	Comparator	Registry family	Period	Std pop	IRR [95% CI]	Main analysis
Shoemaker 2018 [37]	Age <50	Hispanic	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2004-2013	2000 US	0.74 [0.74, 0.75]	yes (representative)
Xu 2024 [22]	Age <50	Hispanic	non-Hispanic White (NHW)	SEER-national	2000-2019	2000 US	0.76 [0.75, 0.77]	no (overlaps representative)
Ihenacho 2023 [45]	Age <50	Japanese	non-Hispanic White (NHW)	Hawaii Tumor Registry (SEER)	2010-2014	2000 US	1.479 [1.252, 1.748]	yes (representative)
Ihenacho 2023 [45]	Age <50	Native Hawaiian	non-Hispanic White (NHW)	Hawaii Tumor Registry (SEER)	2010-2014	2000 US	0.858 [0.710, 1.036]	yes (representative)
Ellington 2022 [25]	Overall (aggregate)	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2018	2000 US	0.683 [0.642, 0.726]	no (AI/AN undercount)
Xie 2022 [49]	Overall (aggregate)	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	1999-2017	2000 US	0.718 [0.707, 0.728]	no (overlaps representative)
Eheman 2009 [40]	Overall (aggregate)	American Indian and Alaska Native (AI/AN)	White (not NH-stratified)	USCS(NPCR+SEER ~99%)	1999-2004	2000 US	0.630 [0.622, 0.639]	no (overlaps representative)
Gopalani 2020 [44]	Overall (aggregate)	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	1999-2015	2000 US	0.560 [0.550, 0.570]	no (overlaps representative)
Zhang 2025 [16]	Overall (aggregate)	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	SEER-national	2017-2021	2000 US	0.81 [0.77, 0.85]	no (overlaps representative)
Du 2022 [30]	Overall (aggregate)	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	SEER-national	2000-2018	2000 US	0.676 [0.659, 0.695]	no (overlaps representative)
Gomez 2026 [31]	Overall (aggregate)	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	SEER-national	2018-2022	2000 US	0.708 [0.677, 0.739]	no (overlaps representative)
Kong 2020 [34]	Overall (aggregate)	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	SEER-national	2010-2015	2000 US	0.82 [0.81, 0.83]	no (overlaps representative)
Hendrick 2021 [3]	Overall (aggregate)	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	SEER-national	2014-2017	2000 US	0.58 [0.52, 0.65]	no (overlaps representative)
SEER22 [64]	Overall (aggregate)	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	SEER-national	2019-2023	2000 US	0.846 [0.808, 0.887]	no (registry-direct anchor)
Wingo 2008 [21]	Overall (aggregate)	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	IHS-PRCDA	1999-2004	2000 US	0.63 [0.61, 0.66]	no (overlaps representative)
Melkonian 2019 [53]	Overall (aggregate)	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	IHS-PRCDA	2010-2015	2000 US	0.87	yes (representative)
Melkonian 2022 [50]	Overall (aggregate)	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	IHS-PRCDA (urban UIHO subset)	2008-2017	2000 US	0.57	no (overlaps representative)
Wright 2022 [51]	Overall (aggregate)	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	State: Massachusetts	2005-2015	2000 US	0.563 [0.33, 0.963]	no (overlaps representative)
Krieger 2018 [54]	Overall (aggregate)	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	State: Massachusetts	2010-2014	2000 US	0.455 [0.292, 0.709]	no (overlaps representative)
Ellington 2022 [25]	Overall (aggregate)	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2018	2000 US	0.769 [0.752, 0.788]	yes (representative)
Xie 2022 [49]	Overall (aggregate)	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	1999-2017	2000 US	0.694 [0.690, 0.698]	no (overlaps representative)
Eheman 2009 [40]	Overall (aggregate)	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	White (not NH-stratified)	USCS(NPCR+SEER ~99%)	1999-2004	2000 US	0.475 [0.458, 0.492]	no (overlaps representative)
Zhang 2025 [16]	Overall (aggregate)	Asian American, Native Hawaiian, and	non-Hispanic White (NHW)	SEER-national	2017-2021	2000 US	0.79 [0.79, 0.8]	no (overlaps representative)

Study	Dimension	Group	Comparator	Registry family	Period	Std pop	IRR [95% CI]	Main analysis
		Pacific Islander (AANHPI)						
Du 2022 [30]	Overall (aggregate)	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	SEER-national	2000-2018	2000 US	0.742 [0.737, 0.747]	no (overlaps representative)
Gomez 2026 [31]	Overall (aggregate)	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	SEER-national	2018-2022	2000 US	0.809 [0.801, 0.817]	no (overlaps representative)
Kong 2020 [34]	Overall (aggregate)	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	SEER-national	2010-2015	2000 US	0.90 [0.89, 0.92]	no (overlaps representative)
Hendrick 2021 [3]	Overall (aggregate)	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	SEER-national	2014-2017	2000 US	0.92 [0.9, 0.95]	no (overlaps representative)
SEER22 [64]	Overall (aggregate)	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	SEER-national	2019-2023	2000 US	0.823 [0.815, 0.831]	no (registry-direct anchor)
Wright 2022 [51]	Overall (aggregate)	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	State: Massachusetts	2005-2015	2000 US	0.567 [0.526, 0.612]	no (overlaps representative)
Krieger 2018 [54]	Overall (aggregate)	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	State: Massachusetts	2010-2014	2000 US	0.642 [0.597, 0.69]	no (overlaps representative)
McCracken 2007 [62]	Overall (aggregate)	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	California-CCR	2000-2002	2000 US	0.588	no (overlaps representative)
Ellington 2022 [25]	Overall (aggregate)	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2018	2000 US	0.933 [0.920, 0.946]	yes (representative)
Cronin 2012 [26]	Overall (aggregate)	non-Hispanic Black (NHB)	White (not NH-stratified)	USCS(NPCR+SEER ~99%)	2005-2009	2000 US	0.957 [0.951, 0.963]	no (overlaps representative)
Xie 2022 [49]	Overall (aggregate)	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	1999-2017	2000 US	0.940 [0.938, 0.943]	no (overlaps representative)
Sung 2020 [52]	Overall (aggregate)	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2010-2016	2000 US	0.98 [0.97, 0.98]	no (overlaps representative)
Richardson 2016 [36]	Overall (aggregate)	non-Hispanic Black (NHB)	White (not NH-stratified)	USCS(NPCR+SEER ~99%)	2009-2013	2000 US	0.983 [0.977, 0.989]	no (overlaps representative)
Eheman 2009 [40]	Overall (aggregate)	non-Hispanic Black (NHB)	White (not NH-stratified)	USCS(NPCR+SEER ~99%)	1999-2004	2000 US	0.826 [0.820, 0.832]	no (overlaps representative)
Davis Lynn 2025 [15]	Overall (aggregate)	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER-national	2013-2015	Segi world	0.971 [0.956, 0.987]	no (overlaps representative)
Zhang 2025 [16]	Overall (aggregate)	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER-national	2017-2021	2000 US	0.93 [0.92, 0.94]	no (overlaps representative)
Brinton 2008 [28]	Overall (aggregate)	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER-national	1992-2004	2000 US	0.85 [0.84, 0.86]	no (overlaps representative)
Du 2022 [30]	Overall (aggregate)	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER-national	2000-2018	2000 US	0.937 [0.931, 0.943]	no (overlaps representative)

Study	Dimension	Group	Comparator	Registry family	Period	Std pop	IRR [95% CI]	Main analysis
Gomez 2026 [31]	Overall (aggregate)	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER-national	2018-2022	2000 US	0.941 [0.932, 0.949]	no (overlaps representative)
Kong 2020 [34]	Overall (aggregate)	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER-national	2010-2015	2000 US	0.96 [0.95, 0.98]	no (overlaps representative)
Hendrick 2021 [3]	Overall (aggregate)	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER-national	2014-2017	2000 US	0.97 [0.95, 1.0]	no (overlaps representative)
SEER22 [64]	Overall (aggregate)	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER-national	2019-2023	2000 US	0.937 [0.929, 0.945]	no (registry-direct anchor)
Zahnd 2019 [18]	Overall (aggregate)	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	Multi-state registry subset (regional)	2012-2014	2000 US	1.07 [1.04, 1.11]	no (overlaps representative)
Wright 2022 [51]	Overall (aggregate)	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	State: Massachusetts	2005-2015	2000 US	0.812 [0.772, 0.855]	no (overlaps representative)
Krieger 2018 [54]	Overall (aggregate)	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	State: Massachusetts	2010-2014	2000 US	0.807 [0.762, 0.855]	no (overlaps representative)
Liu 2015 [43]	Overall (aggregate)	non-Hispanic Black (NHB)	White (not NH-stratified)	State: Texas	1995-2011	2000 US	0.969 [0.934, 1.005]	no (overlaps representative)
Liu 2012 [19]	Overall (aggregate)	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	California-CCR: LA County	2007	2000 US	0.859	no (overlaps representative)
Ellington 2022 [25]	Overall (aggregate)	Hispanic	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2018	2000 US	0.718 [0.707, 0.731]	yes (representative)
Xie 2022 [49]	Overall (aggregate)	Hispanic	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	1999-2017	2000 US	0.710 [0.707, 0.713]	no (overlaps representative)
Carozza 2006 [56]	Overall (aggregate)	Hispanic	non-Hispanic White (NHW)	NAACCR/CiNA (~93%)	1995-2000	2000 US	0.643 [0.637, 0.650]	no (overlaps representative)
Zhang 2025 [16]	Overall (aggregate)	Hispanic	non-Hispanic White (NHW)	SEER-national	2017-2021	2000 US	0.73 [0.72, 0.73]	no (overlaps representative)
Du 2022 [30]	Overall (aggregate)	Hispanic	non-Hispanic White (NHW)	SEER-national	2000-2018	2000 US	0.7 [0.696, 0.705]	no (overlaps representative)
Gomez 2026 [31]	Overall (aggregate)	Hispanic	non-Hispanic White (NHW)	SEER-national	2018-2022	2000 US	0.746 [0.74, 0.751]	no (overlaps representative)
Kong 2020 [34]	Overall (aggregate)	Hispanic	non-Hispanic White (NHW)	SEER-national	2010-2015	2000 US	0.79 [0.75, 0.83]	no (overlaps representative)
Hendrick 2021 [3]	Overall (aggregate)	Hispanic	non-Hispanic White (NHW)	SEER-national	2014-2017	2000 US	0.75 [0.73, 0.76]	no (overlaps representative)
SEER22 [64]	Overall (aggregate)	Hispanic	non-Hispanic White (NHW)	SEER-national	2019-2023	2000 US	0.763 [0.757, 0.769]	no (registry-direct anchor)
Zahnd 2019 [18]	Overall (aggregate)	Hispanic	non-Hispanic White (NHW)	Multi-state registry subset (regional)	2012-2014	2000 US	0.78 [0.69, 0.88]	no (overlaps representative)
Wright 2022 [51]	Overall (aggregate)	Hispanic	non-Hispanic White (NHW)	State: Massachusetts	2005-2015	2000 US	0.571 [0.529, 0.617]	no (overlaps representative)
Krieger 2018 [54]	Overall (aggregate)	Hispanic	non-Hispanic White (NHW)	State: Massachusetts	2010-2014	2000 US	0.617 [0.578, 0.66]	no (overlaps representative)
Wilkinson 2002 [63]	Overall (aggregate)	Hispanic	non-Hispanic White (NHW)	State: Florida	1990-1998	1970 US	0.651 [0.624, 0.679]	no (overlaps representative)
Keegan 2010 [9]	Overall (aggregate)	Hispanic	non-Hispanic White (NHW)	California-CCR	1988-2004	2000 US	0.623 [0.616, 0.630]	no (overlaps representative)
Liu 2012 [19]	Overall (aggregate)	Hispanic	non-Hispanic White (NHW)	California-CCR: LA County	2007	2000 US	0.532	no (overlaps representative)
Gomez 2026 [31]	AANHPI subgroup	Asian American (aggregate)	non-Hispanic White (NHW)	SEER-national	2018-2022	2000 US	0.792 [0.784, 0.800]	yes (representative)
Jin 2016 [60]	AANHPI subgroup	Asian American	non-Hispanic White	Multi-state registry	2009-2011	2000 US	0.703 [0.692, 0.715]	no (overlaps

Study	Dimension	Group	Comparator	Registry family	Period	Std pop	IRR [95% CI]	Main analysis
		(aggregate)	(NHW)	subset (regional)				representative)
Gomez 2026 [31]	AANHPI subgroup	Asian Indian/Pakistani	non-Hispanic White (NHW)	SEER-national	2018-2022	2000 US	0.958 [0.937, 0.979]	yes (representative)
Jin 2016 [60]	AANHPI subgroup	Asian Indian/Pakistani	non-Hispanic White (NHW)	Multi-state registry subset (regional)	2009-2011	2000 US	0.791 [0.758, 0.825]	no (overlaps representative)
Goggins 2009 [47]	AANHPI subgroup	Asian Indian/Pakistani	White (not NH-stratified)	SEER multi-registry subset (metropolitan areas)	1988-2004	SIR (indirect, US White reference)	0.61 [0.56, 0.66] (SIR)	no (overlaps representative)
Gomez 2010 [5]	AANHPI subgroup	Asian Indian/Pakistani	non-Hispanic White (NHW)	California-CCR	1988-2004	2000 US	0.527 [0.494, 0.562]	no (overlaps representative)
Kem 2007 [55]	AANHPI subgroup	Cambodian	non-Hispanic White (NHW)	California-CCR	1998-2002	2000 US	0.264 [0.204, 0.341]	yes (representative)
Gomez 2026 [31]	AANHPI subgroup	Chinese	non-Hispanic White (NHW)	SEER-national	2018-2022	2000 US	0.760 [0.745, 0.775]	yes (representative)
Jin 2016 [60]	AANHPI subgroup	Chinese	non-Hispanic White (NHW)	Multi-state registry subset (regional)	2009-2011	2000 US	0.616 [0.596, 0.637]	no (overlaps representative)
Gomez 2010 [5]	AANHPI subgroup	Chinese	non-Hispanic White (NHW)	California-CCR	1988-2004	2000 US	0.503 [0.490, 0.516]	no (overlaps representative)
McCracken 2007 [62]	AANHPI subgroup	Chinese	non-Hispanic White (NHW)	California-CCR	2000-2002	2000 US	0.491	no (overlaps representative)
Liu 2012 [19]	AANHPI subgroup	Chinese	non-Hispanic White (NHW)	California-CCR: LA County	2007	2000 US	0.575	no (overlaps representative)
Gomez 2026 [31]	AANHPI subgroup	Filipina	non-Hispanic White (NHW)	SEER-national	2018-2022	2000 US	0.857 [0.839, 0.874]	yes (representative)
Jin 2016 [60]	AANHPI subgroup	Filipina	non-Hispanic White (NHW)	Multi-state registry subset (regional)	2009-2011	2000 US	0.828 [0.803, 0.854]	no (overlaps representative)
Gomez 2010 [5]	AANHPI subgroup	Filipina	non-Hispanic White (NHW)	California-CCR	1988-2004	2000 US	0.687 [0.671, 0.704]	no (overlaps representative)
McCracken 2007 [62]	AANHPI subgroup	Filipina	non-Hispanic White (NHW)	California-CCR	2000-2002	2000 US	0.670	no (overlaps representative)
Gomez 2026 [31]	AANHPI subgroup	Guamanian/Chamorro/Samoan	non-Hispanic White (NHW)	SEER-national	2018-2022	2000 US	0.723 [0.641, 0.817]	yes (representative)
Mills 2005 [6]	AANHPI subgroup	Hmong	non-Hispanic White (NHW)	California-CCR	1988-2000	2000 US	0.164 [0.120, 0.224]	yes (representative)
Gomez 2026 [31]	AANHPI subgroup	Japanese	non-Hispanic White (NHW)	SEER-national	2018-2022	2000 US	1.045 [1.009, 1.083]	yes (representative)
Jin 2016 [60]	AANHPI subgroup	Japanese	non-Hispanic White (NHW)	Multi-state registry subset (regional)	2009-2011	2000 US	0.951 [0.908, 0.996]	no (overlaps representative)
Gomez 2010 [5]	AANHPI subgroup	Japanese	non-Hispanic White (NHW)	California-CCR	1988-2004	2000 US	0.702 [0.679, 0.725]	no (overlaps representative)
McCracken 2007 [62]	AANHPI subgroup	Japanese	non-Hispanic White (NHW)	California-CCR	2000-2002	2000 US	0.672	no (overlaps representative)
Gomez 2026 [31]	AANHPI subgroup	Korean	non-Hispanic White (NHW)	SEER-national	2018-2022	2000 US	0.677 [0.654, 0.701]	yes (representative)
Jin 2016 [60]	AANHPI subgroup	Korean	non-Hispanic White (NHW)	Multi-state registry subset (regional)	2009-2011	2000 US	0.562 [0.532, 0.595]	no (overlaps representative)
Gomez 2010 [5]	AANHPI subgroup	Korean	non-Hispanic White (NHW)	California-CCR	1988-2004	2000 US	0.317 [0.300, 0.335]	no (overlaps representative)
Gomez 2003 [57]	AANHPI subgroup	Korean	non-Hispanic White (NHW)	California-CCR	1988-1992	1970 world	0.254 [0.224, 0.288]	no (overlaps representative)

Study	Dimension	Group	Comparator	Registry family	Period	Std pop	IRR [95% CI]	Main analysis
McCracken 2007 [62]	AANHPI subgroup	Korean	non-Hispanic White (NHW)	California-CCR	2000-2002	2000 US	0.332	no (overlaps representative)
Liu 2012 [19]	AANHPI subgroup	Korean	non-Hispanic White (NHW)	California-CCR: LA County	2007	2000 US	0.558	no (overlaps representative)
Gomez 2026 [31]	AANHPI subgroup	Laotian/Kampuchean	non-Hispanic White (NHW)	SEER-national	2018-2022	2000 US	0.388 [0.350, 0.430]	yes (representative)
Gomez 2026 [31]	AANHPI subgroup	Native Hawaiian	non-Hispanic White (NHW)	SEER-national	2018-2022	2000 US	1.270 [1.213, 1.330]	yes (representative)
Miller 2008 [61]	AANHPI subgroup	Native Hawaiian	non-Hispanic White (NHW)	Hawaii Tumor Registry (SEER)	1998-2002	2000 US	1.211 [1.123, 1.305]	no (overlaps representative)
Gomez 2026 [31]	AANHPI subgroup	Native Hawaiian and Pacific Islander (NHPI)	non-Hispanic White (NHW)	SEER-national	2018-2022	2000 US	1.212 [1.168, 1.258]	yes (representative)
Gomez 2026 [31]	AANHPI subgroup	Vietnamese	non-Hispanic White (NHW)	SEER-national	2018-2022	2000 US	0.586 [0.566, 0.608]	yes (representative)
Jin 2016 [60]	AANHPI subgroup	Vietnamese	non-Hispanic White (NHW)	Multi-state registry subset (regional)	2009-2011	2000 US	0.537 [0.506, 0.571]	no (overlaps representative)
Gomez 2010 [5]	AANHPI subgroup	Vietnamese	non-Hispanic White (NHW)	California-CCR	1988-2004	2000 US	0.410 [0.389, 0.433]	no (overlaps representative)
McCracken 2007 [62]	AANHPI subgroup	Vietnamese	non-Hispanic White (NHW)	California-CCR	2000-2002	2000 US	0.363	no (overlaps representative)
Nasseri 2009 [46]	Middle Eastern	Middle Eastern	non-Hispanic White (NHW)	California-CCR	1988-2004	2000 US	0.86 [0.84, 0.88]	yes (representative)
Gleason 2012 [4]	ER+/PR+	non-Hispanic Black (NHB)	White (not NH-stratified)	SEER-national	2004-2008	2000 US	1.459 [1.267, 1.679]	yes (representative)
Gleason 2012 [4]	ER+/PR-	non-Hispanic Black (NHB)	White (not NH-stratified)	SEER-national	2004-2008	2000 US	0.974 [0.925, 1.027]	yes (representative)
Davis Lynn 2025 [15]	HR-	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER-national	2013-2015	Segi world	1.796 [1.740, 1.853]	yes (representative)
Gleason 2012 [4]	HR-	non-Hispanic Black (NHB)	White (not NH-stratified)	SEER-national	2004-2008	2000 US	1.600 [1.548, 1.652]	no (overlaps representative)
Kong 2020 [34]	HR-/HER2+	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	SEER-national	2010-2015	2000 US	1.04 [0.82, 1.31]	yes (representative)
Gomez 2017 [35]	HR-/HER2+	Asian American (aggregate)	non-Hispanic White (NHW)	California-CCR	2009-2013	2000 US	1.21 [1.12, 1.30]	yes (representative)
Howlader 2014 [2]	HR-/HER2+	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	SEER-national	2010	2000 US	1.17 [1.02, 1.35]	no (overlaps representative)
Kong 2020 [34]	HR-/HER2+	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	SEER-national	2010-2015	2000 US	1.41 [1.33, 1.49]	yes (representative)
Sung 2020 [52]	HR-/HER2+	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2010-2016	2000 US	1.29 [1.26, 1.32]	yes (representative)
Howlader 2014 [2]	HR-/HER2+	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER-national	2010	2000 US	1.33 [1.17, 1.52]	no (overlaps representative)
Kong 2020 [34]	HR-/HER2+	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER-national	2010-2015	2000 US	1.46 [1.38, 1.54]	no (overlaps representative)
Zahnd 2019 [18]	HR-/HER2+	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	Multi-state registry subset (regional)	2012-2014	2000 US	1.49 [1.28, 1.71]	no (overlaps representative)
Howlader 2014 [2]	HR-/HER2+	Hispanic	non-Hispanic White	SEER-national	2010	2000 US	0.84 [0.73, 0.96]	no (overlaps

Study	Dimension	Group	Comparator	Registry family	Period	Std pop	IRR [95% CI]	Main analysis
			(NHW)					representative)
Kong 2020 [34]	HR−/HER2+	Hispanic	non-Hispanic White (NHW)	SEER-national	2010-2015	2000 US	1.05 [0.99, 1.11]	yes (representative)
Loo 2019 [24]	HR−/HER2+	Japanese	White (not NH-stratified)	Hawaii Tumor Registry (SEER)	2010-2013	2000 US	0.88 [0.54, 1.44]	yes (representative)
Loo 2019 [24]	HR−/HER2+	Native Hawaiian	White (not NH-stratified)	Hawaii Tumor Registry (SEER)	2010-2013	2000 US	1.19 [0.76, 1.88]	yes (representative)
Davis Lynn 2025 [15]	HR+	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER-national	2013-2015	Segi world	0.820 [0.805, 0.836]	yes (representative)
Gleason 2012 [4]	HR+	non-Hispanic Black (NHB)	White (not NH-stratified)	SEER-national	2004-2008	2000 US	0.703 [0.685, 0.722]	no (overlaps representative)
Kong 2020 [34]	HR+/HER2−	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	SEER-national	2010-2015	2000 US	0.74 [0.69, 0.79]	yes (representative)
Howlader 2014 [2]	HR+/HER2−	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	SEER-national	2010	2000 US	0.689 [0.662, 0.717]	no (overlaps representative)
Kong 2020 [34]	HR+/HER2−	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	SEER-national	2010-2015	2000 US	0.87 [0.85, 0.88]	yes (representative)
Sung 2020 [52]	HR+/HER2−	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2010-2016	2000 US	0.79 [0.78, 0.79]	yes (representative)
Howlader 2014 [2]	HR+/HER2−	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER-national	2010	2000 US	0.739 [0.712, 0.768]	no (overlaps representative)
Kong 2020 [34]	HR+/HER2−	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER-national	2010-2015	2000 US	0.86 [0.84, 0.87]	no (overlaps representative)
Zahnd 2019 [18]	HR+/HER2−	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	Multi-state registry subset (regional)	2012-2014	2000 US	0.87 [0.83, 0.91]	no (overlaps representative)
Lund 2010 [38]	HR+/HER2−	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER: Metropolitan Atlanta	2003-2004	2000 US	0.783	no (overlaps representative)
Loo 2019 [24]	HR+/HER2−	Chinese	White (not NH-stratified)	Hawaii Tumor Registry (SEER)	2010-2013	2000 US	0.58 [0.49, 0.70]	yes (representative)
Loo 2019 [24]	HR+/HER2−	Filipina	White (not NH-stratified)	Hawaii Tumor Registry (SEER)	2010-2013	2000 US	0.64 [0.57, 0.73]	yes (representative)
Howlader 2014 [2]	HR+/HER2−	Hispanic	non-Hispanic White (NHW)	SEER-national	2010	2000 US	0.604 [0.581, 0.627]	no (overlaps representative)
Kong 2020 [34]	HR+/HER2−	Hispanic	non-Hispanic White (NHW)	SEER-national	2010-2015	2000 US	0.78 [0.76, 0.79]	yes (representative)
Zahnd 2019 [18]	HR+/HER2−	Hispanic	non-Hispanic White (NHW)	Multi-state registry subset (regional)	2012-2014	2000 US	0.82 [0.70, 0.95]	no (overlaps representative)
Loo 2019 [24]	HR+/HER2−	Japanese	White (not NH-stratified)	Hawaii Tumor Registry (SEER)	2010-2013	2000 US	1.03 [0.93, 1.15]	yes (representative)
Loo 2019 [24]	HR+/HER2−	Native Hawaiian	White (not NH-stratified)	Hawaii Tumor Registry (SEER)	2010-2013	2000 US	1.12 [1.01, 1.25]	yes (representative)
Kong 2020 [34]	HR+/HER2+	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	SEER-national	2010-2015	2000 US	0.94 [0.81, 1.09]	yes (representative)
Howlader 2014 [2]	HR+/HER2+	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	SEER-national	2010	2000 US	0.850 [0.768, 0.941]	no (overlaps representative)
Kong 2020 [34]	HR+/HER2+	Asian American,	non-Hispanic White	SEER-national	2010-2015	2000 US	1.04 [1.00, 1.08]	yes (representative)

Study	Dimension	Group	Comparator	Registry family	Period	Std pop	IRR [95% CI]	Main analysis
		Native Hawaiian, and Pacific Islander (AANHPI)	(NHW)					
Sung 2020 [52]	HR+/HER2+	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2010-2016	2000 US	1.01 [1.00, 1.03]	yes (representative)
Howlader 2014 [2]	HR+/HER2+	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER-national	2010	2000 US	1.004 [0.916, 1.101]	no (overlaps representative)
Kong 2020 [34]	HR+/HER2+	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER-national	2010-2015	2000 US	1.12 [1.08, 1.16]	no (overlaps representative)
Zahnd 2019 [18]	HR+/HER2+	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	Multi-state registry subset (regional)	2012-2014	2000 US	1.00 [0.90, 1.11]	no (overlaps representative)
Loo 2019 [24]	HR+/HER2+	Chinese	White (not NH-stratified)	Hawaii Tumor Registry (SEER)	2010-2013	2000 US	0.72 [0.42, 1.24]	yes (representative)
Loo 2019 [24]	HR+/HER2+	Filipina	White (not NH-stratified)	Hawaii Tumor Registry (SEER)	2010-2013	2000 US	0.88 [0.61, 1.26]	yes (representative)
Howlader 2014 [2]	HR+/HER2+	Hispanic	non-Hispanic White (NHW)	SEER-national	2010	2000 US	0.687 [0.625, 0.756]	no (overlaps representative)
Kong 2020 [34]	HR+/HER2+	Hispanic	non-Hispanic White (NHW)	SEER-national	2010-2015	2000 US	0.91 [0.88, 0.94]	yes (representative)
Zahnd 2019 [18]	HR+/HER2+	Hispanic	non-Hispanic White (NHW)	Multi-state registry subset (regional)	2012-2014	2000 US	0.74 [0.49, 1.06]	no (overlaps representative)
Loo 2019 [24]	HR+/HER2+	Japanese	White (not NH-stratified)	Hawaii Tumor Registry (SEER)	2010-2013	2000 US	1.03 [0.73, 1.47]	yes (representative)
Loo 2019 [24]	HR+/HER2+	Native Hawaiian	White (not NH-stratified)	Hawaii Tumor Registry (SEER)	2010-2013	2000 US	1.34 [0.96, 1.88]	yes (representative)
Sung 2023 [23]	TNBC	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2015-2019	2000 US	0.860 [0.800, 0.930]	yes (representative)
Zhang 2022 [29]	TNBC	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2019	2000 US	0.734	no (overlaps representative)
Kong 2020 [34]	TNBC	American Indian and Alaska Native (AI/AN)	non-Hispanic White (NHW)	SEER-national	2010-2015	2000 US	0.89 [0.76, 1.04]	no (overlaps representative)
Gomez 2017 [35]	TNBC	Asian American (aggregate)	non-Hispanic White (NHW)	California-CCR	2009-2013	2000 US	0.61 [0.58, 0.65]	yes (representative)
Sung 2023 [23]	TNBC	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2015-2019	2000 US	0.700 [0.680, 0.720]	yes (representative)
Zhang 2022 [29]	TNBC	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2019	2000 US	0.742	no (overlaps representative)
Howlader 2014 [2]	TNBC	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	SEER-national	2010	2000 US	0.640 [0.572, 0.716]	no (overlaps representative)
Kong 2020 [34]	TNBC	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	non-Hispanic White (NHW)	SEER-national	2010-2015	2000 US	0.79 [0.75, 0.83]	no (overlaps representative)
Sung 2023 [23]	TNBC	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2015-2019	2000 US	1.950 [1.930, 1.980]	yes (representative)
Zhang 2022 [29]	TNBC	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2019	2000 US	1.953	no (overlaps representative)

Study	Dimension	Group	Comparator	Registry family	Period	Std pop	IRR [95% CI]	Main analysis
Sung 2020 [52]	TNBC	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2010-2016	2000 US	1.93 [1.90, 1.95]	no (overlaps representative)
Howlader 2014 [2]	TNBC	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER-national	2010	2000 US	1.817 [1.697, 1.946]	no (overlaps representative)
Gomez 2026 [31]	TNBC	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER-national	2018-2022	2000 US	2.0 [1.958, 2.043]	no (overlaps representative)
Kong 2020 [34]	TNBC	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER-national	2010-2015	2000 US	2.07 [2.01, 2.14]	no (overlaps representative)
Zahnd 2019 [18]	TNBC	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	Multi-state registry subset (regional)	2012-2014	2000 US	2.10 [1.94, 2.27]	no (overlaps representative)
Hossain 2019 [59]	TNBC	non-Hispanic Black (NHB)	White (not NH-stratified)	State: Louisiana	2010-2012	not stated (model-adjusted)	2.21 [1.96, 2.48]	no (overlaps representative)
Amirikia 2011 [39]	TNBC	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	California-CCR	1988-2006	2000 US	1.873	no (overlaps representative)
Lund 2010 [38]	TNBC	non-Hispanic Black (NHB)	non-Hispanic White (NHW)	SEER: Metropolitan Atlanta	2003-2004	2000 US	1.871	no (overlaps representative)
Loo 2019 [24]	TNBC	Chinese	White (not NH-stratified)	Hawaii Tumor Registry (SEER)	2010-2013	2000 US	0.53 [0.31, 0.93]	yes (representative)
Loo 2019 [24]	TNBC	Filipina	White (not NH-stratified)	Hawaii Tumor Registry (SEER)	2010-2013	2000 US	0.84 [0.60, 1.18]	yes (representative)
Sung 2023 [23]	TNBC	Hispanic	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2015-2019	2000 US	0.860 [0.840, 0.880]	yes (representative)
Zhang 2022 [29]	TNBC	Hispanic	non-Hispanic White (NHW)	USCS(NPCR+SEER ~99%)	2019	2000 US	0.898	no (overlaps representative)
Howlader 2014 [2]	TNBC	Hispanic	non-Hispanic White (NHW)	SEER-national	2010	2000 US	0.824 [0.756, 0.897]	no (overlaps representative)
Kong 2020 [34]	TNBC	Hispanic	non-Hispanic White (NHW)	SEER-national	2010-2015	2000 US	0.94 [0.91, 0.98]	no (overlaps representative)
Zahnd 2019 [18]	TNBC	Hispanic	non-Hispanic White (NHW)	Multi-state registry subset (regional)	2012-2014	2000 US	0.79 [0.53, 1.13]	no (overlaps representative)
Amirikia 2011 [39]	TNBC	Hispanic	non-Hispanic White (NHW)	California-CCR	1988-2006	2000 US	0.810	no (overlaps representative)
Loo 2019 [24]	TNBC	Japanese	White (not NH-stratified)	Hawaii Tumor Registry (SEER)	2010-2013	2000 US	1.08 [0.79, 1.46]	yes (representative)
Loo 2019 [24]	TNBC	Native Hawaiian	White (not NH-stratified)	Hawaii Tumor Registry (SEER)	2010-2013	2000 US	0.86 [0.60, 1.22]	yes (representative)

Note. This table lists every extracted estimate, not only the ones carried into the main analysis; the “Main analysis” column says which role each plays. Of the 222 rows, 85 are the representative selected for one analytic cell (group × dimension) — the broadest-coverage, most recent source with an appropriate population definition and standardization, a population-based estimate rather than a pooled one — and the remaining 137 are overlaps and demoted or anchor records. Excluding the 4 registry-direct anchor rows, which are reference values rather than screened studies, leaves the 218 extracted estimates reported in the text. Overlapping estimates from nested registries (county ⊂ state ⊂ SEER; SEER and NPCR feed USCS) are not independent and are retained only for the sensitivity re-selection.

Study: the author and year, followed by the study's number in the reference list.

Effect measure: an incidence rate ratio unless the estimate is marked (SIR), a standardized incidence ratio standardized indirectly to the reference population. Both SIRs are overlapping estimates, so every representative — and so every estimate in the main text — is a rate ratio.

Registry family: SEER-national = the pooled SEER 9/13/17/18/21/22 registries; USCS = United States Cancer Statistics, which pools SEER and NPCR to cover nearly the entire U.S. population; NAACCR = the North American Association of Central Cancer Registries; NPCR = the National Program of Cancer Registries; IHS-PRCDA = an Indian Health Service Purchased/Referred Care Delivery Area linkage that improves ascertainment of AI/AN cases; ANTR = the Alaska Native

Tumor Registry; California-CCR = the California Cancer Registry; and “State: X” = a single state central registry (for example, Texas, Louisiana, or Massachusetts).

Comparator: the reference is shown as each source defined it — non-Hispanic White (NHW) where the source stratified by Hispanic origin, or “White (not NH-stratified)” otherwise (concentrated in the receptor-defined subtypes and AI/AN comparisons). 31 of the 222 rows use an unstratified White reference; 19 of those are the representative for their cell, which is the count the main text reports, and the other 12 are overlapping estimates listed here only. The reference and minority rates come from the same source, period, and standard population, so each IRR is internally valid even where the reference is unstratified White; Supplementary Table 6c restricts to NHW-comparator estimates.

Standard population: age-standardized to the 2000 U.S. standard unless the “Std pop” column shows otherwise. Each rate ratio is formed within one study, so its numerator and denominator share that study’s standard population, period, and region and the ratio is internally consistent. The ratio does not, however, cancel the standard population — the age-specific rates of two groups differ in shape, so a different set of standard weights would shift their standardized rates, and their ratio, by different amounts — nor does it make estimates from different studies mutually comparable, because their standard populations, periods, and regions still differ.

Main-analysis codes: “yes (representative)” = carried into the main analysis as the group’s benchmark; “no (overlaps representative)” = an overlapping estimate for the same cell, used only when the representative is re-selected (Table 6); “no (AI/AN undercount)” = an unlinked-registry AI/AN estimate demoted in favour of the IHS-linked representative; “no (registry-direct anchor)” = the SEER-Explorer reference value, not a screened study; “no (external comparator)” = borrows an out-of-paper SEER-Explorer NHW rate, kept only as an overlap record; “no (no usable IRR)” = the source named the subgroup but reported no rate or ratio.

Supplementary Table 5. Risk of bias (JBI checklist for studies reporting prevalence/incidence data)

[illegible]

Study	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	RoB
Melkonian 2022	Y	Y	Y	Y	Y	Y	Y	N	Moderate
Wright 2022	Y	Y	Y	Y	U	Y	Y	Y	Low
Sung 2020	Y	Y	Y	Y	Y	Y	Y	Y	Low
Melkonian 2019	Y	Y	Y	Y	Y	Y	Y	N	Moderate
Krieger 2018	Y	Y	Y	Y	U	Y	Y	Y	Low
Pinheiro 2009	Y	Y	Y	Y	Y	Y	Y	Y	Low
Kem 2007	Y	Y	Y	Y	Y	Y	N	Y	Moderate
Carozza 2006	Y	Y	Y	Y	Y	Y	Y	Y	Low
Gomez 2003	Y	Y	Y	Y	Y	Y	Y	Y	Low
Melkonian 2021	Y	Y	Y	Y	Y	Y	Y	Y	Low
Hossain 2019	Y	Y	Y	Y	U	Y	Y	U	Moderate
Jin 2016	Y	Y	Y	Y	Y	Y	Y	Y	Low
Miller 2008	Y	Y	Y	Y	Y	Y	Y	Y	Low
McCracken 2007	Y	Y	Y	Y	Y	Y	Y	N	Moderate
Wilkinson 2002	Y	Y	Y	Y	Y	Y	Y	Y	Low

Note. Each item is rated Y (Yes), N (No), or U (Unclear); overall risk of bias is Low, Moderate, or High. Item 9 (survey response rate) was not applicable to any study — census-like registry ascertainment has no survey response rate — so it is omitted here and the rating uses the eight applicable items (Q1–Q8).

JBI items: Q1 sample frame appropriate to the target population (defined population-based registry); Q2 appropriate sampling (registry ascertains all diagnosed cases — census-like); Q3 adequate sample size for a stable age-adjusted rate; Q4 study subjects and setting described in detail; Q5 sufficient coverage of the identified population (registry completeness); Q6 valid identification of the condition (invasive breast cancer via registry/pathology record linkage); Q7 condition measured in a standard, reliable way for all participants, including race/ethnicity ascertainment (surname recognition or a known AI/AN undercount = No); Q8 appropriate statistical analysis (age-standardized to a stated standard population with a reported or correctly computed variance).

Overall risk of bias (over the eight applicable items; Q9 = NA excluded): Low = at most one item rated No and both key items (Q7 measurement, Q8 analysis) rated Yes; High = three or more items rated No; Moderate otherwise—so a single No or Unclear on either key item gives at least a Moderate rating. Each study is appraised as the source of the estimates taken from it here rather than as a whole publication, and the items divide accordingly: Q1–Q7 describe the study and its registry (sample frame, ascertainment, case numbers, description, coverage, case identification, and race/ethnicity measurement) and are judged at that level, while Q8 depends on the estimates themselves and is judged on those extracted for this review, a variance counting as available when the source printed an interval or when one could be recovered from what the source reports. A study whose extracted estimates carry no interval therefore rates No on Q8 whether or not the source printed the ratio itself; that rating reflects incomplete reporting of uncertainty, not evidence that the estimate is biased. Of 58 studies assessed, 41 were Low, 17 Moderate, and 0 High; the Moderate ratings arise chiefly where an estimate was reported as a point value without a variance (Q8) or where race/ethnicity ascertainment was limited (Q7).

Supplementary Table 6. Sensitivity analyses

Table 6a. Low-risk-of-bias only (Moderate/High-RoB studies dropped): 54 of 85 cells unchanged, 19 changed, 12 dropped.

Dimension	Group	Main analysis: IRR [95% CI] — study	Under restriction: IRR [95% CI] — study	Status
AI/AN by region	AI/AN (East)	0.61 — Melkonian 2019	0.53 [0.45, 0.63] — Wingo 2008	changed
AI/AN by region	AI/AN (Northern Plains)	1.05 — Melkonian 2019	0.89 [0.81, 0.97] — Wingo 2008	changed
AI/AN by region	AI/AN (Pacific Coast)	0.93 — Melkonian 2019	0.52 [0.48, 0.57] — Wingo 2008	changed
AI/AN by region	AI/AN (Southwest)	0.57 — Melkonian 2019	0.41 [0.37, 0.44] — Wingo 2008	changed
Overall (aggregate)	American Indian and Alaska Native (AI/AN)	0.87 — Melkonian 2019	0.63 [0.61, 0.66] — Wingo 2008	changed
AANHPI subgroup	Asian American (aggregate)	0.792 [0.784, 0.800] — Gomez 2026	0.703 [0.692, 0.715] — Jin 2016	changed
AANHPI subgroup	Asian Indian/Pakistani	0.958 [0.937, 0.979] — Gomez 2026	0.791 [0.758, 0.825] — Jin 2016	changed
AANHPI subgroup	Chinese	0.760 [0.745, 0.775] — Gomez 2026	0.616 [0.596, 0.637] — Jin 2016	changed
AANHPI subgroup	Filipina	0.857 [0.839, 0.874] — Gomez 2026	0.828 [0.803, 0.854] — Jin 2016	changed
AANHPI subgroup	Japanese	1.045 [1.009, 1.083] — Gomez 2026	0.951 [0.908, 0.996] — Jin 2016	changed
AANHPI subgroup	Korean	0.677 [0.654, 0.701] — Gomez 2026	0.562 [0.532, 0.595] — Jin 2016	changed
AANHPI subgroup	Native Hawaiian	1.270 [1.213, 1.330] — Gomez 2026	1.211 [1.123, 1.305] — Miller 2008	changed
AANHPI subgroup	Vietnamese	0.586 [0.566, 0.608] — Gomez 2026	0.537 [0.506, 0.571] — Jin 2016	changed
HR–/HER2+	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	1.41 [1.33, 1.49] — Kong 2020	1.17 [1.02, 1.35] — Howlader 2014	changed
HR–/HER2+	Hispanic	1.05 [0.99, 1.11] — Kong 2020	0.84 [0.73, 0.96] — Howlader 2014	changed
HR+/HER2–	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	0.87 [0.85, 0.88] — Kong 2020	0.689 [0.662, 0.717] — Howlader 2014	changed
HR+/HER2–	Hispanic	0.78 [0.76, 0.79] — Kong 2020	0.604 [0.581, 0.627] — Howlader 2014	changed
HR+/HER2+	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	1.04 [1.00, 1.08] — Kong 2020	0.850 [0.768, 0.941] — Howlader 2014	changed
HR+/HER2+	Hispanic	0.91 [0.88, 0.94] — Kong 2020	0.687 [0.625, 0.756] — Howlader 2014	changed
Age ≥50	American Indian and Alaska Native (AI/AN)	0.647 — Harper 2009	not available under this restriction	dropped
Age ≥50	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	0.659 — Harper 2009	not available under this restriction	dropped
Age ≥50	non-Hispanic Black (NHB)	0.914 — Harper 2009	not available under this restriction	dropped
Age ≥50	Hispanic	0.660 — Harper 2009	not available under this restriction	dropped
AANHPI subgroup	Cambodian	0.264 [0.204, 0.341] — Kem 2007	not available under this restriction	dropped
AANHPI subgroup	Guamanian/Chamorro/Samoan	0.723 [0.641, 0.817] — Gomez 2026	not available under this restriction	dropped
AANHPI subgroup	Laotian/Kampuchean	0.388 [0.350, 0.430] — Gomez 2026	not available under this restriction	dropped
AANHPI subgroup	Native Hawaiian and Pacific Islander (NHPI)	1.212 [1.168, 1.258] — Gomez 2026	not available under this restriction	dropped
Middle Eastern	Middle Eastern	0.86 [0.84, 0.88] — Nasseri 2009	not available under this restriction	dropped
HR–/HER2+	American Indian and Alaska Native (AI/AN)	1.04 [0.82, 1.31] — Kong 2020	not available under this restriction	dropped
HR+/HER2–	American Indian and Alaska Native (AI/AN)	0.74 [0.69, 0.79] — Kong 2020	not available under this restriction	dropped
HR+/HER2+	American Indian and Alaska Native (AI/AN)	0.94 [0.81, 1.09] — Kong 2020	not available under this restriction	dropped

Table 6b. Directly-reported IRR only (estimates the review computed from rates, and the two SIRs, are dropped): 35 unchanged, 4 changed, 46 dropped — most disaggregated and subtype cells rely on computed rates, because registries publish rates rather than ratios.

Dimension	Group	Main analysis: IRR [95% CI] — study	Under restriction: IRR [95% CI] — study	Status
Age <40	non-Hispanic Black (NHB)	1.183 [1.142, 1.226] — Anderson 2008	1.16 [1.1, 1.23] — Baquet 2008	changed
Overall (aggregate)	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	0.769 [0.752, 0.788] — Ellington 2022	0.79 [0.79, 0.8] — Zhang 2025	changed
Overall (aggregate)	non-Hispanic Black (NHB)	0.933 [0.920, 0.946] — Ellington 2022	0.98 [0.97, 0.98] — Sung 2020	changed
Overall (aggregate)	Hispanic	0.718 [0.707, 0.731] — Ellington 2022	0.73 [0.72, 0.73] — Zhang 2025	changed
Hispanic by origin	Cuban	0.769 [0.688, 0.860] — Pinheiro 2009	not available under this restriction	dropped
Hispanic by origin	Mexican	0.512 [0.382, 0.686] — Pinheiro 2009	not available under this restriction	dropped
Hispanic by origin	New Latino	0.697 [0.629, 0.771] — Pinheiro 2009	not available under this restriction	dropped
Hispanic by origin	Puerto Rican	0.833 [0.738, 0.939] — Pinheiro 2009	not available under this restriction	dropped
Age ≥40	non-Hispanic Black (NHB)	0.851 [0.841, 0.862] — Anderson 2008	not available under this restriction	dropped
Age ≥50	American Indian and Alaska Native (AI/AN)	0.647 — Harper 2009	not available under this restriction	dropped
Age ≥50	Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	0.659 — Harper 2009	not available under this restriction	dropped
Age ≥50	non-Hispanic Black (NHB)	0.914 — Harper 2009	not available under this restriction	dropped
Age ≥50	Filipina	0.774 [0.700, 0.855] — Ihenacho 2023	not available under this restriction	dropped
Age ≥50	Hispanic	0.660 — Harper 2009	not available under this restriction	dropped
Age ≥50	Japanese	1.064 [0.978, 1.157] — Ihenacho 2023	not available under this restriction	dropped
Age ≥50	Native Hawaiian	1.366 [1.246, 1.499] — Ihenacho 2023	not available under this restriction	dropped
Age <50	Filipina	0.768 [0.633, 0.932] — Ihenacho 2023	not available under this restriction	dropped
Age <50	Japanese	1.479 [1.252, 1.748] — Ihenacho 2023	not available under this restriction	dropped
Age <50	Native Hawaiian	0.858 [0.710, 1.036] — Ihenacho 2023	not available under this restriction	dropped
AANHPI subgroup	Asian American (aggregate)	0.792 [0.784, 0.800] — Gomez 2026	not available under this restriction	dropped
AANHPI subgroup	Asian Indian/Pakistani	0.958 [0.937, 0.979] — Gomez 2026	not available under this restriction	dropped
AANHPI subgroup	Cambodian	0.264 [0.204, 0.341] — Kem 2007	not available under this restriction	dropped
AANHPI subgroup	Chinese	0.760 [0.745, 0.775] — Gomez 2026	not available under this restriction	dropped
AANHPI subgroup	Filipina	0.857 [0.839, 0.874] — Gomez 2026	not available under this restriction	dropped
AANHPI subgroup	Guamanian/Chamorro/Samoan	0.723 [0.641, 0.817] — Gomez 2026	not available under this restriction	dropped
AANHPI subgroup	Hmong	0.164 [0.120, 0.224] — Mills 2005	not available under this restriction	dropped
AANHPI subgroup	Japanese	1.045 [1.009, 1.083] — Gomez 2026	not available under this restriction	dropped
AANHPI subgroup	Korean	0.677 [0.654, 0.701] — Gomez 2026	not available under this restriction	dropped
AANHPI subgroup	Laotian/Kampuchean	0.388 [0.350, 0.430] — Gomez 2026	not available under this restriction	dropped
AANHPI subgroup	Native Hawaiian	1.270 [1.213, 1.330] — Gomez 2026	not available under this restriction	dropped
AANHPI subgroup	Native Hawaiian and Pacific Islander (NHPI)	1.212 [1.168, 1.258] — Gomez 2026	not available under this restriction	dropped
AANHPI subgroup	Vietnamese	0.586 [0.566, 0.608] — Gomez 2026	not available under this restriction	dropped
ER−/PR+	non-Hispanic Black (NHB)	1.459 [1.267, 1.679] — Gleason 2012	not available under this restriction	dropped
ER+/PR−	non-Hispanic Black (NHB)	0.974 [0.925, 1.027] — Gleason 2012	not available under this restriction	dropped
HR−	non-Hispanic Black (NHB)	1.796 [1.740, 1.853] — Davis Lynn 2025	not available under this restriction	dropped
HR−/HER2+	Japanese	0.88 [0.54, 1.44] — Loo 2019	not available under this restriction	dropped

Dimension	Group	Main analysis: IRR [95% CI] — study	Under restriction: IRR [95% CI] — study	Status
HR–/HER2+	Native Hawaiian	1.19 [0.76, 1.88] — Loo 2019	not available under this restriction	dropped
HR+	non-Hispanic Black (NHB)	0.820 [0.805, 0.836] — Davis Lynn 2025	not available under this restriction	dropped
HR+/HER2–	Chinese	0.58 [0.49, 0.70] — Loo 2019	not available under this restriction	dropped
HR+/HER2–	Filipina	0.64 [0.57, 0.73] — Loo 2019	not available under this restriction	dropped
HR+/HER2–	Japanese	1.03 [0.93, 1.15] — Loo 2019	not available under this restriction	dropped
HR+/HER2–	Native Hawaiian	1.12 [1.01, 1.25] — Loo 2019	not available under this restriction	dropped
HR+/HER2+	Chinese	0.72 [0.42, 1.24] — Loo 2019	not available under this restriction	dropped
HR+/HER2+	Filipina	0.88 [0.61, 1.26] — Loo 2019	not available under this restriction	dropped
HR+/HER2+	Japanese	1.03 [0.73, 1.47] — Loo 2019	not available under this restriction	dropped
HR+/HER2+	Native Hawaiian	1.34 [0.96, 1.88] — Loo 2019	not available under this restriction	dropped
TNBC	Chinese	0.53 [0.31, 0.93] — Loo 2019	not available under this restriction	dropped
TNBC	Filipina	0.84 [0.60, 1.18] — Loo 2019	not available under this restriction	dropped
TNBC	Japanese	1.08 [0.79, 1.46] — Loo 2019	not available under this restriction	dropped
TNBC	Native Hawaiian	0.86 [0.60, 1.22] — Loo 2019	not available under this restriction	dropped

Table 6c. Non-Hispanic White comparator only (unstratified-White comparators dropped): 66 of 85 cells unchanged, 1 changed, 18 dropped.

Dimension	Group	Main analysis: IRR [95% CI] — study	Under restriction: IRR [95% CI] — study	Status
AI/AN by region	Alaska Native	1.09 [0.99, 1.21] — Nash 2019	1.25 [1.11, 1.41] — Melkonian 2021	changed
Age ≥40	non-Hispanic Black (NHB)	0.851 [0.841, 0.862] — Anderson 2008	not available under this restriction	dropped
Age <40	non-Hispanic Black (NHB)	1.183 [1.142, 1.226] — Anderson 2008	not available under this restriction	dropped
ER–/PR+	non-Hispanic Black (NHB)	1.459 [1.267, 1.679] — Gleason 2012	not available under this restriction	dropped
ER+/PR–	non-Hispanic Black (NHB)	0.974 [0.925, 1.027] — Gleason 2012	not available under this restriction	dropped
HR–/HER2+	Japanese	0.88 [0.54, 1.44] — Loo 2019	not available under this restriction	dropped
HR–/HER2+	Native Hawaiian	1.19 [0.76, 1.88] — Loo 2019	not available under this restriction	dropped
HR+/HER2–	Chinese	0.58 [0.49, 0.70] — Loo 2019	not available under this restriction	dropped
HR+/HER2–	Filipina	0.64 [0.57, 0.73] — Loo 2019	not available under this restriction	dropped
HR+/HER2–	Japanese	1.03 [0.93, 1.15] — Loo 2019	not available under this restriction	dropped
HR+/HER2–	Native Hawaiian	1.12 [1.01, 1.25] — Loo 2019	not available under this restriction	dropped
HR+/HER2+	Chinese	0.72 [0.42, 1.24] — Loo 2019	not available under this restriction	dropped
HR+/HER2+	Filipina	0.88 [0.61, 1.26] — Loo 2019	not available under this restriction	dropped
HR+/HER2+	Japanese	1.03 [0.73, 1.47] — Loo 2019	not available under this restriction	dropped
HR+/HER2+	Native Hawaiian	1.34 [0.96, 1.88] — Loo 2019	not available under this restriction	dropped
TNBC	Chinese	0.53 [0.31, 0.93] — Loo 2019	not available under this restriction	dropped
TNBC	Filipina	0.84 [0.60, 1.18] — Loo 2019	not available under this restriction	dropped
TNBC	Japanese	1.08 [0.79, 1.46] — Loo 2019	not available under this restriction	dropped
TNBC	Native Hawaiian	0.86 [0.60, 1.22] — Loo 2019	not available under this restriction	dropped

Table 6d. 2000 U.S. standard population only (estimates on other standard populations dropped): 83 of 85 cells unchanged, 2 changed, 0 dropped.

Dimension	Group	Main analysis: IRR [95% CI] — study	Under restriction: IRR [95% CI] — study	Status
HR–	non-Hispanic Black (NHB)	1.796 [1.740, 1.853] — Davis Lynn 2025	1.600 [1.548, 1.652] — Gleason 2012	changed
HR+	non-Hispanic Black (NHB)	0.820 [0.805, 0.836] — Davis Lynn 2025	0.703 [0.685, 0.722] — Gleason 2012	changed

The first value column is the representative estimate in the main analysis; the second is the representative re-selected after applying the restriction (low-risk-of-bias only in Table 6a; directly reported IRR only in Table 6b, where the two SIRs are excluded because they standardize indirectly to the reference

population; NHW-comparator only in Table 6c; 2000 U.S. standard population only in Table 6d). Each value is followed by the study that supplied it, so a changed cell shows both the study that is replaced and the study that replaces it. Status: unchanged = the same study remains the representative; changed = a different study becomes the representative; dropped = no estimate in that cell met the restriction. Every changed and dropped cell is listed; the cells not listed were unchanged. A dropped cell is not evidence that the main estimate is robust — the restriction leaves that cell untested, so dropped cells should be read as unexamined under that restriction rather than as confirmation.

Supplementary Note 1. Provenance of estimates and derivation log

Transparency record. Every estimate whose IRR was not printed as-is by the study is listed here with how it was derived, so each is reproducible. Estimates reported by the source as a ratio (‘directly-reported-IRR’ / ‘-SIR’) are taken verbatim and are not in this log.

1. Age-standardization from age-specific rates

- Howlader 2014 (Suppl Table 3). The supplement reports 5-year age-specific incidence rates (ages 20–85+) by race × subtype, not a single age-adjusted rate. Each race's 14 age-specific rates were standardized to the 2000 US standard population (Census P25-1130 weights for the 20–85+ groups): $\text{std_rate} = \sum(\text{rate}_i \cdot w_i) / \sum w_i$. $\text{IRR} = \text{minority_std} / \text{NHW_std}$.
 - Alignment verified against printed anchors (HR+/HER2– NHW 45–49 = 128.2, 75–79 = 342.7; TNBC NHW 35–39 = 9.7).
 - 95% CIs computed by variance propagation. Suppl Table 3 reports a 95% CI for every age-specific rate. $\text{SE per band} = (\text{hi} - \text{lo}) / (2 \cdot 1.96)$; $\text{Var}(\text{std_rate}) = \sum(w_i / \sum w_i)^2 \cdot \text{SE}_i^2$; $\text{SE}(\log \text{IRR}) = \sqrt{(\text{SE}_{\text{min}} / \text{std_min})^2 + (\text{SE}_{\text{nhw}} / \text{std_nhw})^2}$. Provenance = ‘computed-from-rates-with-CI’ (e.g., TNBC Black 1.82 [1.70, 1.95], HR+/HER2– AANHPI 0.69 [0.66, 0.72]).
 - IRR is essentially unaffected by the 20+ vs all-age standard (ages 0–19 contribute ≈0 to both numerator and denominator).
- Amirikia 2011 (Amirikia 2011, CCR). Table 2 gives TNBC rates by 5 age groups (0–39, 40–49, 50–59, 60–74, ≥75) per race; standardized to the 2000 US standard (P25-1130 group weights) → all-ages TNBC rates NHB 23.6, NHW 12.6, Hispanic 10.2 (reproduces our values). $\text{IRR} = \text{minority} / \text{NHW}$ (point; source gives per-age CIs only): Black 1.87, Hispanic 0.81.

2. IRR computed from race-specific rates ± CIs

$\text{IRR} = \text{minority_rate} / \text{NHW_rate}$. With both rate CIs: $\text{SE}(\log \text{IRR}) = \sqrt{\text{SE}_{\text{min}}^2 + \text{SE}_{\text{nhw}}^2}$, $\text{SE}(\log \text{rate}) = (\ln \text{hi} - \ln \text{lo}) / (2 \cdot 1.96)$; provenance ‘computed-from-rates-with-CI’. Rate without CI → point IRR only; provenance ‘computed-from-rates’ (or ‘directly-reported-rate’ where the source printed the rate). The records in each class:

- with CI: Davis Lynn 2025, Loo 2019, Gleason 2012, Du 2022, Anderson 2008, Keegan 2010, Richardson 2016, Ehemann 2009, Liu 2015, Ihenacho 2023, Xie 2022, Wright 2022, Krieger 2018, Carozza 2006, Gomez 2003, Miller 2008.
- Ehemann 2009 (Ehemann 2009, NPCR+SEER, 44 states + DC, 1999–2004). Table 5 prints age-standardized (2000 US) total invasive lobular/ductal/mixed breast IR with 95% CIs by race, White 108.1 [107.9–108.3]: Black 89.3/108.1 = 0.826 [0.820, 0.832], Asian/PI 51.3/108.1 = 0.475 [0.458, 0.492], AI/AN 68.1/108.1 = 0.630 [0.622, 0.639] (delta method). Comparator is unstratified White; a national estimate over the ductal/lobular/mixed subset (~87% of invasive breast), each entering its aggregate-vs-NHW cell as a sensitivity overlap of the USCS-national representative (Ellington 2022), which it does not displace (lower coverage, unstratified comparator, older period).
- Liu 2015 (Liu 2015, Texas Cancer Registry, 1995–2011). Table II prints age-adjusted (2000 US) breast IR with 95% CIs, White 136.14 [131.91–140.37], Black 131.90 [129.40–134.40]: Black/White = 0.969 [0.934, 1.005] (delta method). Comparator is unstratified White; the paper's own stage/grade-adjusted incidence ratio is 0.98 [0.97, 0.99]. A single-state estimate entering the Black aggregate-vs-NHW cell as a sensitivity overlap of the national representative.
- Krieger 2018 (Krieger 2018, Massachusetts Cancer Registry, 2010–2014). Table 2 prints age-standardized (2000 US) breast incidence rates with 95% CIs by race, NHW 140.8 [139.0–142.7]: NHB 113.6/140.8 = 0.807 [0.762, 0.855], Hispanic 86.9/140.8 = 0.617 [0.578, 0.660], Asian/PI 90.4/140.8 = 0.642 [0.597, 0.690], AI/AN 64.1/140.8 = 0.455 [0.292, 0.709] (delta method from the reported rate CIs). A single-state estimate; each enters its aggregate-vs-NHW cell as a sensitivity overlap of the USCS-national representative.
- Wright 2022 (Wright/Krieger 2022, Massachusetts Cancer Registry, 28 HOLC municipalities, 2005–2015). Table 3 prints age-standardized (2000 US) breast IR by race, NHW 146.56 [143.91–149.25]: NHB 119.06/146.56 = 0.812 [0.772, 0.855], Hispanic 83.71 = 0.571 [0.529, 0.617], Asian/PI 83.15 = 0.567 [0.526, 0.612], AI/AN 82.57 = 0.563 [0.330, 0.963] (delta method). A single-state estimate (overlaps the national aggregate reps and the other Massachusetts study, Krieger 2018).
- Du 2022 (Du 2022, SEER 18, 2000–2018). The table prints age-adjusted rates (2000 US std) by race with 95% CIs, referenced to Asian/PI (not NHW). We recompute each group's IRR vs NHW from the same-source rates and rate CIs by the delta method (minority IR / NHW IR 190.4 [190.0–190.8]): Asian/PI 141.3/190.4 = 0.742 [0.737, 0.747], NHB 178.4/190.4 = 0.937 [0.931, 0.943], AI/AN 128.8/190.4 = 0.676 [0.659, 0.695], Hispanic 133.3/190.4 = 0.700 [0.696, 0.705]. SEER-18 overall estimates; each enters its aggregate-vs-NHW cell as a sensitivity overlap of the USCS-national representative.

- Loo 2019 (Loo 2019, Hawaii HTR). The source printed IRRs with implausibly narrow CIs (e.g. Native Hawaiian HR+/HER2+ 1.35 [1.347-1.351]); its Table 1 rate CIs, however, are correct for the sample sizes. We therefore discard the source IRR CIs and recompute each subtype IRR and its CI from the Table 1 age-adjusted rates and rate CIs by the delta method (minority IR / White IR; comparator = unstratified Whites, †). Example: Chinese HR+/HER2– 66.8 [56.9-78.1] / White 114.3 [106.4-122.6] = 0.58 [0.49, 0.70] (source gave 0.58 [0.46-0.53], which excluded its own point estimate). Verified: recomputed points reproduce the source IRRs (e.g. Japanese HR+/HER2– 118.1/114.3 = 1.03).
- point only (no CI): Liu 2012, Lund 2010, Watanabe-Galloway 2015, Harper 2009, McCracken 2007.
- Handled in other sections: §1 (Howlader 2014), §4 (Gomez 2026, Pinheiro 2009, Nash 2022), §4b (Wilkinson 2002), and the Poisson-SE records in §3.
- Davis Lynn 2025 rates use the Segi world 1960 standard (ages 20-74), not 2000 US — the IRR remains comparable (a ratio); the absolute scale differs (flagged in-row).
- Anderson 2008, Richardson 2016 use "White"/"Black" not NH-stratified (Richardson 2016 White incl. Hispanic); Anderson 2008 is the crossover from printed age-specific rates.
- Carozza 2006 (Carozza 2006, NAACCR/CiNA, 15 registries, 1995-2000). The table reports female invasive breast rates with standard errors (2000 US std): Latina 89.2 (SE 0.44), non-Hispanic White 138.7 (SE 0.21). IRR = 89.2/138.7 = 0.643, reproducing the paper's printed SIR of 0.64; 95% CI (0.637, 0.650) by the delta method from the reported rate SEs (rate CIs = rate ± 1.96·SE). An all-Hispanic multi-state estimate; enters the overall Hispanic aggregate cell as a sensitivity overlap of the USCS-national representative (Ellington 2022, 0.72), and overlaps Wilkinson 2002 (Miami Hispanic).
- Watanabe-Galloway 2015 (Watanabe-Galloway 2015, NE/ND/SD state registries, 2002-2009). Table 2 prints all-ages age-standardized (2000 US) breast rates: AI/AN 134.6, NHW 149.3 (printed RR 0.9). IRR = 134.6/149.3 = 0.902, point estimate only (no rate confidence intervals reported). These NE/ND/SD registries are unlinked and undercount AI/AN. The IHS-PRCDA Melkonian 2019 study (Melkonian 2019) also reports a Northern-Plains breast estimate (RR 1.05), so Watanabe-Galloway 2015 is not the only Northern-Plains source; because the IHS-linked source corrects the AI/AN undercount, Melkonian 2019 (1.05) is the Northern-Plains representative and Watanabe-Galloway 2015 (0.902, unlinked, undercount-prone) is demoted to an overlap/sensitivity row.

3. Poisson SE from case counts

Where a study reports rates and annual case counts but no ratio CI, $SE(\log IRR) = \sqrt{1/D_{\min} + 1/D_{\text{nhw}}}$ (D = cases); provenance `computed-from-rates-Poisson-SE`. Limitation: these are age-standardized rates, so a total-count Poisson SE does not fully capture the age-specific weighting of the variance; it is a deliberate approximation used only where the source reported no rate SE or interval. Where a source did report rate CIs the CI is instead propagated by the delta method (§1, §4, §4b) — e.g. Cronin 2012 (Cronin) was moved to a delta-method CI from its printed rate CIs (Black 116.9 [116.2-117.6], White 122.1 [121.8-122.3]), giving 0.957 [0.951, 0.963] in place of the earlier total-count Poisson interval.

- Ellington 2022 (USCS). Aggregate age-adjusted rates for women aged ≥20 (USCS, ~99% coverage): Black 174.0, Hispanic 134.0, Asian/PI 143.5, AI/AN 127.3, vs NHW 186.5; the paper reports rates, not ratios. IRR = minority / 186.5, with the CI from the annual case counts (e.g., Black 0.933 [0.920, 0.946], Hispanic 0.718 [0.707, 0.731]). This supplies three of the four aggregate overall estimates.
- Mills 2005 (Hmong-California). Hmong female breast AAIR 23.8 (39 cases) vs NHW 145.5; only the Hmong case count (39) is available, so $SE(\log IRR) = 1/\sqrt{39} = 0.160$ with the large NHW denominator treated as fixed ($1/D_{\text{nhw}} \sim 0$). IRR 0.164 (0.120-0.224). Period mismatch (Hmong 1988-2000 vs NHW 1995-1999) flagged in-row.
- Kem 2007 (Cambodian CA+PugetSound). Author-reported Cambodian/ White ratio 0.26 (surname-method "high rate" 41.0 / NHW 155.5); the high/low rate bracket 34.8-41.0 is an ethnic-identification uncertainty range, not a 95% CI, so $SE(\log IRR) = 1/\sqrt{58} = 0.131$ from the 58 Cambodian cases. IRR 0.264 (0.204-0.341).

4. NHW comparator reported apart from the minority rates (same-paper companion table or supplement; Nash 2022 the one external comparator)

- Gomez 2026. Disaggregated AANHPI rates are in the main Table (2018-2022, SEER-21); the in-paper NHW comparator is eTable 3 (Supplement 1): Non-Hispanic White 139.5 (139.0-140.0), same registry (SEER-21), same period (2018-2022), same standard (2000 US). IRRs = subgroup rate / 139.5 with 95% CIs propagated from both rate CIs.
- Pinheiro 2009. Table 4 gives origin-specific female-breast rates with 95% CIs (Cuban 108.0, Mexican 71.9, New Latino 97.8, Puerto Rican 116.9), and the non-Hispanic White Florida comparator is reported in Table 2: 140.4 (137.6-143.2), same registry (Florida, 1999-2001), same standard (2000 US). IRR = origin rate / 140.4, with the 95% CI propagated from both rate CIs by the delta method (e.g., Cuban 0.769 [0.688, 0.860]). These reproduce the authors' printed rounded rate ratios (0.8/0.5/0.7/0.8).

- Nash 2022 (Alaska Native). Paper is ANTR-only (Alaska Native people), no in-paper NHW. AN female breast 130.8 (116.7-146.0), 2014-2018, 2000 US std. Paired with the SEER-Explorer NHW observed annual mean for the SAME window (2014-2018 = 137.4, 2000 US std). IRR 0.952 (0.851-1.065). This overlaps Nash 2019 (same ANTR registry, 2009-2014), which reports a directly-reported IRR 1.09 (0.99-1.21) vs an in-paper US White reference; Nash 2019 is the cell representative and Nash 2022 is a sensitivity/overlap estimate (the 0.95 vs 1.09 difference reflects the different period and the external NHW denominator).

4b. IRR CI from reported standard errors of rates

- Wilkinson 2002 (Miami-Dade Hispanic). Table 1 gives invasive breast rates with SEs: Hispanic 81.9 (SE 1.2), NHW 125.8 (SE 2.0), 1990-1998. $IRR = 81.9/125.8 = 0.651$; $SE(\log \text{ rate}) = SE_{\text{rate}}/\text{rate}$; $SE(\log IRR) = \sqrt{(1.2/81.9)^2 + (2.0/125.8)^2} = 0.0216 \rightarrow IRR \text{ 0.651 (0.624-0.679)}$. Regional aggregate Hispanic (Cuban-heavy); enters the overall Hispanic aggregate cell as a sensitivity overlap (with Carozza 2006) of the USCS-national representative (Ellington 2022).

5. Author-reported rate ratios taken as directly reported

- Nasser 2009 (Middle Eastern-California). The authors print the rate ratio and its CI: invasive breast 0.86 (126.2/146.9, 95% CI 0.84-0.88); taken as `directly-reported-IRR`. Source states the US 2000 standard (incidence-table footnote: "Rates are age adjusted to the US 2000 population standards").
- Goggins 2009 (Asian Indian/Pakistani). Table 1 prints the female breast SIR relative to US White = 0.61 (0.56-0.66); `directly-reported-SIR`. Overlaps Gomez 2026 Asian Indian/Pakistani (sensitivity).

6. Direction conversion

- Kong 2020. Subtype IRRs are printed minority-vs-NHW (verified from "In [minority]... higher/lower than NHW") and taken as-is. The overall comparison for Black women is printed the other way round — "non-Hispanic White women was 31.3 ... , which was higher compared with the incidence among Black women (IRR, 1.04; 95% CI, 1.02-1.05)" — whereas the same paragraph gives the paper's other aggregates as minority-vs-NHW (Asian/PI 0.90, AI/AN 0.82, Hispanic 0.79). That row is therefore inverted to Black/NHW 0.96 [0.95, 0.98] so its direction matches both the review's scale and the paper's own other aggregates; provenance `directly-reported-IRR-inverted`.

7. Overlap studies entered for the sensitivity analysis (not representatives)

These five were read from full text and enter the ledger only as overlap/sensitivity rows; each collapses to an existing cell representative (finalize_representatives), so none changes a main-text estimate.

- Gomez 2010 (Gomez 2010, California Cancer Registry, 1988-2004). Table 1 prints age-standardized (2000 US) invasive breast rates: NHW 146.1 (145.5-146.7); Chinese 73.5 (71.6-75.4), Japanese 102.5 (99.3-105.9), Filipina 100.4 (98.1-102.8), Korean 46.3 (43.8-49.0), Vietnamese 59.9 (56.7-63.1), and South Asian [=Asian Indian/Pakistani] 77.0 (72.1-82.1). $IRR = \text{rate}/146.1$ with delta-method CI (e.g. Chinese 0.503, 0.490-0.516; South Asian 0.527, 0.494-0.562). State registry (< national SEER-21); overlaps the SEER-21 subgroup reps (Gomez 2026).
- Jin 2016 (Jin 2016, eight-state SEER+NPCR, 2009-2011). Table 4 (female) prints breast rates with CIs: NHW 134.4 (133.8-135.1); Chinese 82.8, Filipina 111.3, Japanese 127.8, Korean 75.6, South Asian [=Asian Indian/Pakistani] 106.3, Vietnamese 72.2, Asian aggregate 94.5. $IRR = \text{rate}/134.4$, delta-method CI. Eight-state subset (< national SEER-21); overlaps Gomez 2026.
- Melkonian 2022 (Melkonian 2022, urban IHS-linked USCS-AIAD, 2008-2017). Table 2 prints all-ages breast rates for urban populations: AI/AN 74.2, NHW 129.6, RR 0.57. $IRR = 74.2/129.6 = 0.573$. Urban-restricted IHS subset (< full PRCDA); overlaps the AI/AN aggregate representative (Gopalani 2020).
- Amirikia 2011 (Amirikia 2011, California Cancer Registry, 1988-2006). Table 2 prints age-specific triple-negative rates by race (0-39, 40-49, 50-59, 60-74, >=75). These were age-standardized by the review to all ages with 2000 US standard-population weights (0.5697/0.1540/0.1112/0.1048/0.0603): NHB 23.6, NHW 12.6, Hispanic 10.2 per 100,000. $IRR(\text{Black})=1.87$, $IRR(\text{Hispanic})=0.81$. California registry (< national USCS); overlaps the USCS TNBC rep (Sung 2023).
- Zhang 2022 (Zhang 2022, NPCR+SEER/USCS, 2019). The text (Figure 2A) prints the 2019 age-adjusted triple-negative rates by race: NHB 25.0, NHW 12.8, Hispanic 11.5, AI/AN 9.4, AANHPI 9.5 per 100,000. $IRR = \text{rate}/12.8$ (NHB 1.95, Hispanic 0.90, AI/AN 0.73, AANHPI 0.74). Single-year 2019 cross-section; overlaps the USCS TNBC representative (Sung 2023, 2015-2019 mean), which is preferred on the longer span.